

Closure Without Catastrophe: Patient-Flow Exposure and the Mortality Effects of Psychiatric Deinstitutionalization in Brazil

Darcio Genicolo-Martins*

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Abstract

Critics of psychiatric deinstitutionalization warned that closing hospitals would raise patient mortality. Testing this requires knowing who actually depended on a closing hospital, and where patients cross municipal borders for specialized care the distance-based exposure rule standard in the closure literature misidentifies them—misclassifying 293 of the 337 municipality–closure pairs in our sample. We replace it with a patient-flow exposure rule built from 179.4 million Brazilian SUS admissions and estimate effects on 48 PNASH-driven psychiatric closures with a staggered event study, testing pre-trends on data back to 2010. A first stage validates the rule: travel burden falls 7.1 km after closure, twice what distance-based exposure detects. For the feared outcome we find no harm—suicide mortality changes by +0.28 per 100,000 (95% CI $[-0.72, +1.29]$), ruling out the large increases critics predicted, and self-harm is likewise null. We are candid about a limit: preventable hospitalizations were already rising before closure, so we do not read them causally. The catastrophe predicted for Brazil’s psychiatric reform does not appear among the populations that bore the closures—the ones a distance-based analysis would have missed.

Keywords: hospital closures, deinstitutionalization, patient-flow exposure, mortality, suicide, staggered difference-in-differences, SUS, Brazil.

JEL: I18, I14, I11, H75.

*Insper. Email: darcio.g.martins@gmail.com. Working paper, comments welcome. Replication code, pipeline, and cached LLM/CNPJ classifications: <https://github.com/darciogm/bitter-pills/tree/main/paper18-hospital-deserts>.

1 Introduction

Deinstitutionalization—the closing of psychiatric hospitals in favor of community-based care—is among the most contested health reforms of the past half-century. Its critics have long warned that removing inpatient capacity faster than communities can absorb the displaced patients raises mortality, and suicide in particular, among those who relied on the closing facilities (Yoon and Bruckner, 2009; Bastiampillai et al., 2016; Pirkola et al., 2009). Brazil ran this experiment at scale: the Reforma Psiquiátrica (Lei 10.216/2001), enforced through federal PNASH inspection cycles and a chronic real-value erosion of SUS reimbursement, closed specialized psychiatric hospitals across the country during the 2010s. Whether those closures harmed the patients who depended on them is a first-order policy question—and answering it credibly turns on a measurement problem that the closure literature has not confronted.

The problem is who counts as exposed. The canonical closure literature defines exposure by geographic proximity: a population is treated by a closure if the closing hospital was among its nearest providers (Avdic, 2016; Carroll, 2019; Joynt et al., 2015; Gujral and Basu, 2019; Lindrooth et al., 2018). But specialized care has thin local substitutes, and in a single-payer system patients routinely cross municipal borders to reach it—so proximity mismeasures dependence. Consider one closure in our sample: a specialized psychiatric facility in São Paulo state closed in 2016 had redirected 2,331 pre-closure SUS admissions drawn from 20 municipalities, 16 of which lay between 36 and 107 km from the facility and would not have been treated under any standard top-3-nearest distance rule.¹ The choice of exposure measure is not a technicality: it determines whether the population the policy actually touched enters the estimate at all.

We replace proximity with a *patient-flow* exposure rule. Municipality m is exposed to the closure of hospital h if, in the year before closure, admissions from m to h exceeded a 5% share of m ’s total SUS admissions—dependence revealed by where patients actually went, not by how near h happened to be. The rule is built from the bipartite municipality–hospital graph of 179.4 million SUS hospital admissions. Across the 60 exogenous closures, the flow rule and the distance rule disagree on 293 of the 337 municipality–closure pairs that either rule flags: distance *misses* 97 pairs whose patients depended on the closing hospital through specialized referral, and *falsely flags* 196 whose patients never used it. The disagreement is not random—it concentrates in the river-corridor North, where patient flow reaches regional hubs by routes road distance ignores, and the semi-arid Northeast, where the geographically near hospital is bypassed for a more distant referral center (Figure 1).

A first stage confirms the flow rule captures genuine dependence rather than noise. Travel burden—the distance the typical patient travels for inpatient care—falls by 7.1 km after closure in the exposed municipalities, where a distance-based exposure rule applied to the same closures recovers only about half as much. The fall is direct evidence that exposed patients had been traveling

¹CNES 2082683, IBGE municipality of closure 350280; total pre-closure SUS admissions from exposed municipalities = 2,331; median distance from the closing hospital to an exposed municipality = 44 km. Reproducible from `02_data/intermediate/exposure_panel.parquet` in the replication package.

far for care with no local substitute: when the facility closes, their realized travel drops because the long trips it generated cease. (We summarize the broader referral network with a node2vec embedding of the flow graph; it serves as a measurement diagnostic in Appendix C and plays no role in the causal design.)

We estimate effects on the psychiatric closures with a staggered Sun-Abraham event study (Sun and Abraham, 2021), with Callaway and Sant’Anna (2021) and Borusyak et al. (2024) as alternative estimators. The identifying variation is the *timing* of closure, which we anchor to the federal PNASH inspection cycles and the national SUS reimbursement freeze—institution-level forces that operate on a recurring federal calendar rather than on municipality-specific demand or supply shocks (Section 2). Our primary sample is the 48 PNASH-anchored psychiatric closures, exposing 104 municipalities. Mortality is recorded at the municipality of residence. Crucially, the parallel-trends assumption is tested on full pre-periods: we extend the population denominators back to 2010, so the pre-treatment trajectory of every cohort is observed rather than missing—a data gap that, left unfilled, would make the pre-trend test uninformative for the earlier cohorts.

We find no evidence of the feared harm. Among the exposed population, population-weighted suicide mortality changes by +0.28 per 100,000 after closure (95% CI $[-0.72, +1.29]$), with parallel pre-trends that do not reject ($p = 0.26$); self-harm mortality is likewise indistinguishable from zero. The confidence interval rules out the increases above one-quarter of baseline that deinstitutionalization’s critics predicted. We are careful about what this null does and does not say: it excludes a mortality *catastrophe*, not small effects, which the data cannot resolve. The null is robust across exposure samples, across population- and municipality-weighting, across modern staggered estimators, and across the exposure-share threshold.

One outcome we decline to read causally. Preventable hospitalizations (ICSAP), the canonical Brazilian access-to-care marker (Macinko and Harris, 2015; Rocha and Soares, 2010), do rise in exposed municipalities after closure—but they were *already* rising before it. On the full pre-periods, the parallel-trends test rejects sharply: the post-closure increase is the continuation of a pre-existing trend, not a treatment effect. We report the trajectory in full and treat ICSAP as a non-identified outcome rather than launder a pre-trend into a causal claim. This discipline is part of the contribution: a credible null requires being equally honest about the outcomes that are identified and the ones that are not.

The paper makes two contributions. *Methodological*: a portable patient-flow exposure measure for hospital-closure studies in single-payer systems with origin–destination claims data, validated against a mechanical first stage and contrasted with the distance rule it corrects. *Substantive*: a credibly identified mortality null on a globally contested reform, estimated on the population that a distance-based analysis would not have found. Section 2 situates the analysis in the SUS and the Reforma Psiquiátrica. Section 3 documents the data. Section 4 develops the flow exposure measure and the identification strategy. Section 5 reports the first stage, the mortality null, and the ICSAP non-result. Section 6 reports the robustness battery. Section 7 discusses what the null means for

deinstitutionalization policy and the measurement lesson for closure studies.

2 Institutional setting

The policy question—did closing psychiatric hospitals harm the patients who depended on them?—and the measurement problem at its heart both arise from three features of the Brazilian healthcare system. First, SUS regionalization sends patients across municipal and regional boundaries at scale, so geographic proximity is a poor proxy for who depends on a given hospital (2.1). Second, the Reforma Psiquiátrica (Lei 10.216/2001), combined with a chronic real-value erosion of SUS reimbursements, closed specialized psychiatric hospitals on a federally-driven timeline that supplies our identifying variation (2.2). Third, the closure events themselves, recovered from the federal establishment register, form the sample we analyze (2.3).

2.1 SUS regionalization and cross-border patient flow

Brazil’s Sistema Único de Saúde, established by the 1988 Constitution and operationalized by Laws 8.080 and 8.142 (1990), is a publicly financed, universal system covering the entire resident population in principle. Approximately 71% of Brazilians rely on it exclusively and a further 25% use it as a complement to private insurance (Macinko and Harris, 2015). Inpatient care is delivered through a mix of municipal, state, philanthropic, and private contracted providers. The spatial architecture is nested: each of Brazil’s 5,570 municipalities is a management unit; municipalities aggregate into 438 Regiões de Saúde (CIRs) meant to articulate flows across levels of care; and the 27 Federation Units operate or contract the referral hospitals at the apex.

Cross-border flow at scale. The architecture prescribes a regional flow hierarchy; it does not enforce one. Patients cross municipal and CIR boundaries routinely, particularly for specialized care. Across the 179.4 million SUS hospital admissions in our pooled bipartite graph (2010–2024), 33.2% are recorded against a hospital outside the patient’s municipality of residence (volume-weighted); the unweighted municipality-mean outflow share is 70%, reflecting that smaller municipalities depend far more on elsewhere than the national volume average suggests. Cross-border rates exceed 55% in Sergipe, Pernambuco, and the Distrito Federal. These flows are not residual—they are how a large share of the SUS population reaches inpatient care. The consequence for closure analysis is direct: a patient whose routine referral hospital sits across a CIR boundary is recorded as “unexposed” to its closure under a nearest-hospital distance rule, even though that hospital was her actual destination. The distance proxy standard in the closure literature (Aldic, 2016; Carroll, 2019; Joynt et al., 2015; Lindrooth et al., 2018) silently assumes flow aligns with road geography—defensible in systems with thick local capacity (Sweden, much of the NHS), questionable even in the U.S. Medicare context, where Wang et al. (2020) document drift between patient flows and the Hospital Referral Region boundaries the Dartmouth Atlas has used since the early 1990s. Brazil’s 33.2% cross-border share

is larger still, and is precisely where a flow-based exposure correction is most likely to matter.

2.2 The Reforma Psiquiátrica and the source of closure timing

Our closure sample is dominated by an ongoing structural reform of mental-health care, and the reform’s architecture is both the source of our identifying variation and the reason the mortality question is politically charged. Lei 10.216/2001 prescribes deinstitutionalization: the progressive replacement of asylum-style psychiatric hospitals (*hospitais psiquiátricos*) by community outpatient networks (Centros de Atenção Psicossocial, CAPS) and supported residences (Serviços Residenciais Terapêuticos). Critics of the reform have warned throughout its implementation that closing inpatient beds faster than communities build substitutes would leave the most vulnerable patients without care, raising suicide and crisis mortality (Bastiampillai et al., 2016; Yoon and Bruckner, 2009). Whether that fear was realized is the question this paper answers. Implementation proceeded along two reinforcing federal tracks.

Track 1: PNASH inspection cycles. The Programa Nacional de Avaliação dos Serviços Hospitalares (PNASH) audited every accredited psychiatric hospital on cycles in 2002–2003, 2007–2008, 2011–2012, and 2015–2016, scoring infrastructure, staffing, and care processes; facilities below roughly 61% on the composite index were issued an indication for de-accreditation under Portaria SAS/MS 501/2007. The recurring inspection schedule is the critical feature for identification: closure timing is structured by a federal calendar exogenous to municipality-specific demand or supply shocks. Our primary sample anchors to it (closures within ± 1 year of a cycle; Section 4.2).

Track 2: chronic real-value erosion of reimbursement. The SUS daily reimbursement for psychiatric inpatient care was nominally near R\$30 per patient-day from the early 2000s, unadjusted for inflation; over the panel window real per-diem reimbursement fell by roughly half. The squeeze produced institution-level insolvency at scale on a timeline driven by federal reimbursement policy, not by local fiscal capacity or local demand.

Both pressures apply identical federal thresholds and reimbursement schedules across all 5,570 municipalities and generate closure decisions at the institution rather than the municipality level. The closures we study are therefore not induced by municipality-specific demand shocks, local fiscal changes, or local primary-care access; they are induced by federal policy reaching individual institutions. An event-by-event natural-language classification of declared motives (Appendix E) cross-checks this mechanism for each closure; because its external validation is incomplete, we use it only to define robustness sub-samples, not as the primary exogeneity warrant.

2.3 Closure events, 2010–2024

Closure events are identified from the Cadastro Nacional de Estabelecimentos de Saúde (CNES), the federal establishment register. We flag establishments whose reporting ceases before the end of the window; the raw count is dominated by registry artefacts (small clinics, day units, code changes). A multi-stage filter (F1–F5; Appendix A) reduces the set to 60 closures meeting five cumulative criteria: closure in the non-pandemic windows 2012–2018 or 2022–2023; hospital-grade establishment type ($\text{tp_unid} \in \{05, 07, 15, 62\}$); ≥ 30 SUS-contracted beds and ≥ 100 admissions in the pre-closure year; and no $> 50\%$ admission decline in the pre-closure year (ruling out gradual demand-driven exit). Of the 60, 41 are specialized hospitals ($\text{tp_unid} = 07$, predominantly psychiatric), 18 general ($\text{tp_unid} = 05$), and 1 a day unit. The Southeast holds 62% and the Northeast 25%; the densest year is 2012 (11 events), reflecting the early psychiatric-audit cycles.

The analysis sample. The mortality analysis restricts to the psychiatric closures, where the PNASH–reimbursement exogeneity argument is strongest and the deinstitutionalization question is posed. Our primary sample is the 48 psychiatric closures within ± 1 year of a PNASH cycle; the 41 statistically-filtered specialized closures and a broader 60-closure psychiatric set serve as robustness (Section 6). Closure-motive classifications, used to form additional robustness sub-samples, are cross-tabulated against facility type in Appendix E: administrative closures concentrate in specialized hospitals (20 of 30), consistent with the PNASH de-accreditation mechanism.

3 Data

The analysis assembles a municipality-year panel covering the universe of Brazilian municipalities, their public hospital network, inpatient admissions (2010–2024), and mortality (2010–2023). Four primary sources contribute: the federal hospital admissions register (SIH-RD, 3.1), which yields both the patient-flow exposure measure and the travel-burden outcome; the federal vital registration database (SIM, 3.2), which yields the mortality outcomes; the federal establishment register (CNES, 3.3), which yields the closure events; and IBGE auxiliary panels for population and centroids (3.4). Documentary sources used only to form robustness sub-samples are described in Appendix E. Table 1 reports summary statistics by macro-region.

Two patterns matter for the design. First, travel burden is structurally unequal: Norte and Centro-Oeste municipalities average 94–96 km from residence to admitting hospital, more than twice the 42–45 km of the Sudeste and Sul. Second, the outflow share—SUS admissions sent outside the residential municipality—is high everywhere (58–73%), which is what makes a within-municipality, closure-driven design operate on real referral dependence rather than noise. The same cross-region heterogeneity also confounds any cross-sectional comparison of access and health, which is why we identify effects only from within-municipality variation under a closure shock.

Table 1: Summary statistics by Brazilian macro-region, 2015–2023 pooled.

Region	Munis	Pop (med, k)	GDP (med, R\$M)	Travel (km)	ICSAP per 1k	Outflow share
Norte	449	16.8	0.2	95.7	12.6	0.58
Nordeste	1,794	14.4	0.1	69.1	12.4	0.73
Centro-Oeste	466	9.5	0.3	94.3	13.5	0.62
Sudeste	1,668	11.3	0.3	42.3	13.6	0.71
Sul	1,188	7.2	0.3	45.0	18.9	0.73
National	5,565	11.5	0.2	59.9	14.3	0.70

Notes: Cells are municipality-level statistics averaged across 2015–2023. *Pop* is the median municipality population in thousands; *GDP* is the median municipal GDP in millions of R\$ (constant); *Travel* is the average weighted-mean centroid-to-centroid haversine distance from residence to admission hospital; *ICSAP per 1k* is the average rate of primary-care-sensitive hospitalizations per 1,000 inhabitants (Portaria MS 221/2008 list); *Outflow share* is the municipality-mean share of admissions sent outside the residential municipality (the volume-weighted national counterpart, 33.2%, is reported in Section 2.1). Source: SIH-RD 2010–2024; IBGE 2015–2024 population and 2010–2023 municipal GDP panels.

3.1 SIH-RD: admissions, the flow graph, and travel burden

The Sistema de Informações Hospitalares–Reduzido (SIH-RD) is the billing record for every SUS-financed inpatient admission. Each record is one Autorização de Internação Hospitalar (AIH). Our universe spans 2010–2024 at monthly granularity: 179.4 million admissions across municipalities of patient residence (MUNIC_RES) and roughly 7,000 active SUS hospitals (CNES). The register covers the SUS-using population only ($\sim 96\%$ of residents; private inpatient flows are unobserved), so our estimates are identified within that population (Section 7.4).

From this microdata we build the weighted bipartite graph whose edges (m, h, t) count admissions of residents of m at hospital h in year t ; this graph defines the flow exposure shares of equation (1). Two outcomes are derived from it. *Travel burden* is the admission-weighted haversine distance from m ’s centroid to the centroid of the municipality of admission, at the municipality–year level; it is the first-stage outcome that validates the exposure rule (Section 5.1). *ICSAP* is the count of admissions whose primary diagnosis (DIAG_PRINC) falls in the Brazilian primary-care-sensitive list (Portaria MS 221/2008) per 1,000 population, the canonical Brazilian access marker (Macinko and Harris, 2015; Rocha and Soares, 2010); we report it transparently as a non-identified outcome (Section 5.3).

3.2 SIM: psychiatric mortality outcomes

The Sistema de Informação sobre Mortalidade (SIM) records every death in Brazil with cause coded in ICD-10; our window comprises 18.9 million deaths during 2010–2023, attributed to the municipality of *residence* (CODMUNRES). The reform’s critics warned specifically about deaths of discharged or displaced psychiatric patients, so we construct two residence-based mortality outcomes that target that fear directly. *Suicide* mortality counts deaths from intentional self-harm (ICD-10 X60–X84) per 100,000. A broader *self-harm* measure adds deaths of undetermined intent (Y10–Y34),

which absorb the misclassification of suicides as accidents that is common in vital registration. We also track deaths with a mental or behavioural disorder as the underlying cause (F00–F99) as an auxiliary series; because such deaths partly reflect in-facility mortality that mechanically falls when an asylum closes, we do not treat it as a welfare outcome.

Why mortality is recorded at residence, and why that matters. Recording deaths by residence rather than occurrence is essential here: a displaced patient who dies after traveling—or after failing to travel—to care is counted against her home municipality, which is the unit our exposure rule treats. (The occurrence-based attribution that SIM also carries would displace such deaths to the hospital’s municipality, breaking the link to the exposed population; we use the residence field throughout.) Two limits of SIM remain. Ill-defined-cause recording (R96–R99) is higher in the rural North and Northeast (14–22% versus 4–6% in the South), adding noise to mortality rates there; and suicide is known to be under-recorded relative to undetermined-intent deaths, which is why we report the broader self-harm measure alongside. Both are absorbed by municipality fixed effects in the event study to the extent they are time-invariant.

3.3 CNES: establishments, beds, and closure detection

The Cadastro Nacional de Estabelecimentos de Saúde (CNES) is the federal register of every health establishment, updated monthly. The establishment register (`cnes_st`) provides establishment type (`tp_unid`) and municipality; the bed register (`cnes_lt`) provides SUS-contracted beds (`QT_SUS`), used in the closure filter. Closures are identified by the cessation of `cnes_st` reporting for a CNES after a candidate year. The candidate set (3,518 establishments ceasing to report between 2010 and 2023) is dominated by registry artefacts; the cumulative F1–F5 cascade (Appendix A) reduces it to 60 exogenous closures, 41 of them specialized psychiatric hospitals (Section 2.3). The cascade does not eliminate every false positive, but the closure-motive classification of Appendix E provides a cross-check used to form robustness sub-samples.

3.4 IBGE auxiliary: population and centroids

Municipality population denominators come from the IBGE annual estimates (SIDRA table 6579) and the 2022 Census. A point of construction matters for the pre-trend test: the rate outcomes require a population denominator in every panel year, and the early cohorts’ pre-treatment years fall before 2015. We therefore extend the denominators back to 2010—using the IBGE annual estimates for 2011–2014 and a linear back-extrapolation for 2010—so that every cohort’s pre-treatment trajectory is observed rather than missing. Without this backfill the pre-period of the earlier cohorts is empty and the parallel-trends test is uninformative; with it, the test is real, and it is what reveals the pre-existing ICSAP trend (Section 5.3). Municipality centroids (IBGE shapefile, 2022 vintage) enter the haversine distance behind the distance exposure rule E2 and the travel-burden outcome.

The centroid-to-centroid distance approximation. CNES does not record hospital coordinates, so distances are computed centroid-to-centroid between the patient’s municipality and the admitting hospital’s municipality—standard in the Brazilian access literature (Macinko and Harris, 2015; Rocha and Soares, 2010). The approximation understates within-municipality distance in large urban municipalities and near-border cases, attenuating the distance measure; excluding the 27 state capitals (the principal sites of hospital dispersion) changes the first-stage travel ATT by less than 1% (Section 6); a coordinate-level sensitivity for a sub-sample of closures appears in Appendix J.

4 Measuring exposure and identifying closure effects

The analysis rests on a single measurement choice—who counts as exposed to a hospital closure—and a single source of identifying variation—the timing of closure. Section 4.1 defines the patient-flow exposure rule and contrasts it with the distance rule it replaces. Section 4.2 states the exogeneity argument and the three identification assumptions. Section 4.3 gives the estimator. Throughout, the object is the effect of a closure on the populations that depended on the closing hospital; the methodological contribution is getting that population right.

4.1 Patient-flow exposure

The flow rule (E1). For each closing hospital h with closure year $t_h^\star$, let

$$s_{m,h} = \frac{w_{t_h^\star-1}^\star(m, h)}{\sum_{h' \in H_{t_h^\star-1}^\star} w_{t_h^\star-1}^\star(m, h')} \quad (1)$$

denote the share of municipality m ’s SUS admissions in the year preceding closure that were sent to h , where $w_t(m, h)$ counts admissions from residents of m to hospital h in year t , drawn from the bipartite municipality–hospital graph of 179.4 million SUS admissions (Section 3). Municipality m is *flow-exposed* to the closure of h if $s_{m,h} \geq \theta$, with default $\theta = 0.05$. The rule encodes dependence as revealed by care-seeking: a municipality whose residents systematically traveled to h is exposed even when h is distant, and a geographically near hospital that residents never used is correctly excluded. Robustness to a continuous sweep $\theta \in [0.01, 0.20]$ is reported in Section 6.

The distance rule (E2). The comparison is the exposure rule standard in the closure literature: m is *distance-exposed* to h if h is among the three kilometer-nearest hospitals to the centroid of m in the pre-closure year,

$$m \in \text{E2}(h) \iff h \in \operatorname{argmin}_{h' \in H_{t_h^\star-1}^{(3)}} \text{km}(m, h'), \quad (2)$$

with top-3 chosen as a generous catchment so the comparison set is not artificially small.

Where the two rules disagree. The rules are not nested. Across the 60 exogenous closures, of the 337 municipality–closure pairs that either rule flags, only 44 are flagged by both: 97 are flow-only (the patients depended on h but it was not among their nearest hospitals) and 196 are distance-only (the nearest-hospital catchment, but the residents never used h). Distance thus misclassifies 293 of the 337 pairs. The disagreement is geographically systematic (Figure 1): in the river-corridor North, flow reaches regional hubs by routes road distance ignores; in the semi-arid Northeast, the near hospital is bypassed for a more distant referral center; the South, with mature SUS regionalization, shows the smallest divergence. The first-stage travel-burden result (Section 5.1) confirms which rule is right: the flow-exposed municipalities are the ones whose travel actually falls when h closes.

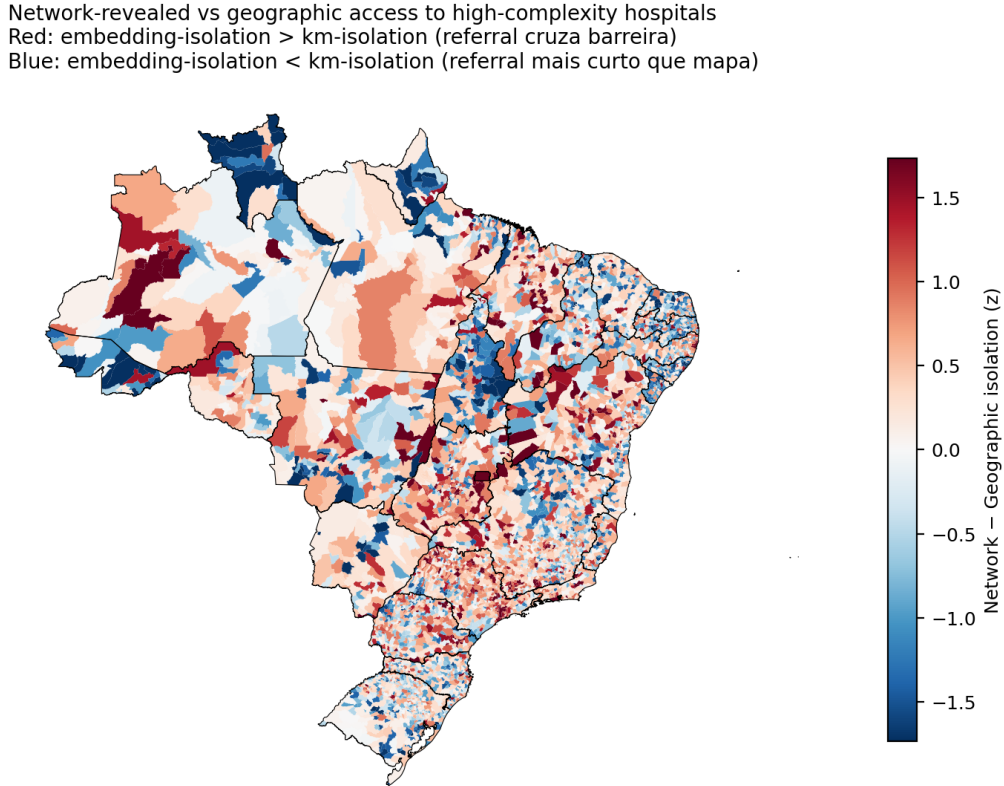


Figure 1: Where flow and distance disagree. Standardized divergence $\Delta_m = z(\text{iso}_m^{\text{flow}}) - z(\text{iso}_m^{\text{km}})$ across 5,565 Brazilian municipalities. Red: the flow measure identifies m as more isolated than distance suggests (patients bypass the near hospital). Blue: less isolated than distance suggests (river/air access not captured by road km). Color bar truncated at $\pm$ the 95th absolute percentile.

The embedding as a measurement diagnostic. We also summarize the referral network with a node2vec embedding of the patient-flow graph, which recovers Brazilian metropolitan and *ribeirinho* geography from co-attendance alone and supplies the divergence measure mapped in Figure 1. The embedding is a descriptive diagnostic, not the treatment assignment: the causal rule is the share-based E1 of equation (1), which is invariant to the embedding and to its random seed.

Construction, stability across seeds, and the cross-sectional finding that the embedding does *not* outperform simple distance or travel-time baselines for pooled mortality prediction are reported in Appendix C; none of it bears on the within-event estimates below.

4.2 Identification: closure exogeneity and the three assumptions

The exogenous shock. The flow share $s_{m,h}$ is not randomly assigned—it is the equilibrium of a referral network shaped by past policy and patient sorting—and we do not treat it as such. The identifying variation is the *closure event* $h \rightarrow \emptyset$ at year t_h^* , and its exogeneity to local healthcare demand rests on institutional timing. The psychiatric closures in our sample were driven by the federal PNASH inspection cycles and the chronic real-value erosion of SUS reimbursement schedules (Section 2)—forces operating on a recurring national calendar rather than on municipality-specific demand or capacity shocks. Our primary sample restricts to the 48 closures falling within ± 1 year of a PNASH cycle, where the institutional timing component is sharpest; the full set of exogenous psychiatric closures (`tp_unid` = 07, $n = 41$ under the statistical filter, $n = 60$ relaxing it) appears as robustness in Section 6.

Three assumptions. (i) *Parallel pre-trends*: in the absence of closure, exposed and never-treated municipalities would have followed the same average trajectory. We test this on pre-period event-study coefficients observed back to 2010 (the population denominators are backfilled to give the early cohorts real pre-periods; Section 3). The assumption holds for the mortality outcomes and the first stage, and fails for ICSAP—which we therefore decline to identify (Section 5.3). (ii) *No anticipation*: closures are quasi-public events with weeks-to-months of notice, not years; we set the anticipation horizon to zero, with pre-period coefficients as the implicit test. (iii) *No spillover*: a closure may redirect patients to a hospital also serving an untreated municipality, biasing its outcome. We bound this by excluding state capitals—the principal referral hubs—in Section 6; the mortality null is unaffected.

Pre-treatment balance. Table 2 reports balance between flow-exposed and never-treated municipalities. Population and GDP are statistically indistinguishable (standardized differences ≤ 0.03 , $p > 0.79$); exposed municipalities have somewhat *lower* pre-treatment travel burden and ICSAP, ruling out selection on bad baseline health; the share of admissions sent outside the residential municipality is mechanically higher in the exposed set, by construction of the flow rule.

4.3 Estimator

We estimate effects with the staggered event study of Sun and Abraham (2021), which is robust to the heterogeneous treatment effects across cohorts that a closure sample spanning 2012–2023 invites. Let i index municipalities (flow-exposed and never-treated controls), t index years in [2010, 2024],

Table 2: Pre-treatment balance (years 2015–2017, the window where SUS pop, GDP, and outcome data are jointly available) between E1-treated municipalities ($n = 130$) and never-treated municipalities ($n = 5435$). Standardized differences computed as $(m_T - m_C)/\sqrt{(s_T^2 + s_C^2)/2}$.

Pre-treatment covariate	Treated mean (sd)	Control mean (sd)	Std. diff.	p -value
log(population)	9.44 (1.44)	9.47 (1.16)	-0.026	0.791
log(municipal GDP)	12.27 (1.73)	12.25 (1.39)	+0.011	0.909
travel burden 2015-17 (km)	48.65 (31.80)	58.38 (44.76)	-0.251	0.001***
ICSAP rate 2015-17 (per 1k)	12.04 (6.89)	15.97 (12.52)	-0.389	0.000***
share of outflow admissions 2015-17	0.79 (0.33)	0.69 (0.32)	+0.326	0.000***

Welch’s t -test. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and g_i the year of the earliest closure to which i is exposed ($g_i = \infty$ for never-treated). With $\text{rel}_{i,t} = t - g_i$,

$$Y_{i,t} = \sum_{e \neq -1} \sum_g \delta_{g,e} \mathbf{1}\{g_i = g\} \mathbf{1}\{\text{rel}_{i,t} = e\} + \alpha_i + \gamma_t + \varepsilon_{i,t}, \quad (3)$$

with municipality fixed effects α_i , year fixed effects γ_t , and reference period $e = -1$. We aggregate the cohort-specific $\hat{\delta}_{g,e}$ into the dynamic trajectory (plotted in the event studies) and the simple post-period average (the ATT reported in tables), following [Sun and Abraham \(2021\)](#). Standard errors are clustered at the municipality level.

Weighting. For the rate outcomes we report population-weighted estimates as primary: the policy-relevant quantity is the effect on the average exposed *person*, and population weights also stabilize the per-capita rates of small municipalities. The unweighted (municipality-weighted) estimate is reported alongside; both deliver the same null (Section 5.2).

Alternative estimators and machinery. The mortality null is reproduced under [Callaway and Sant’Anna \(2021\)](#) and [Borusyak et al. \(2024\)](#) in Section 6. A causal-forest attribution of heterogeneity ([Athey et al., 2019](#)) and a counterfactual graph-perturbation diagnostic, both auxiliary to the main design, are reported in Appendices G and F. A natural-language classification of declared closure motives, used to cross-check the exogeneity filter, is described in Appendix E; its external validation is incomplete, so it informs robustness only. An auxiliary in-hospital cardiac-mortality outcome, computed at the receiving hospital to sidestep the SIM residence-recording issues, is reported in Appendix I.

5 Results

We report three results in sequence. First, a validation: the patient-flow exposure rule captures populations that genuinely depended on the closing hospital, evidenced by a sharp fall in travel burden (5.1). Second, the headline: among those exposed populations, the mortality the reform’s

critics feared did not materialize (5.2). Third, a result we decline to read causally: preventable hospitalizations rise after closure, but they were already rising before it (5.3). The full robustness battery is in Section 6.

5.1 First stage: the flow rule captures real dependence

Before asking what happened to the exposed populations, we verify that the flow exposure rule identifies municipalities that actually relied on the closing hospital rather than ones merely near it. The test is mechanical and sharp: if exposed patients had been traveling far for care that the closing facility uniquely provided, then closure should cut the distance they travel for inpatient care, because the long trips the facility generated cease.

Figure 2 reports the staggered event study for travel burden. The pre-period coefficients are flat—the joint Wald-style test does not reject parallel trends ($p = 0.96$)—and travel burden falls by 7.1 km in the post-period (95% CI $[-11.07, -3.10]$; Table 3, first row). A distance-based exposure rule applied to the same closures recovers only about half this magnitude, because it dilutes the genuinely dependent municipalities with geographically near ones whose residents never used the closing hospital. The fall confirms that flow-exposed municipalities were sending patients on long trips to the facility that closed: these are the populations a closure plausibly affects, and the ones a distance rule misclassifies.

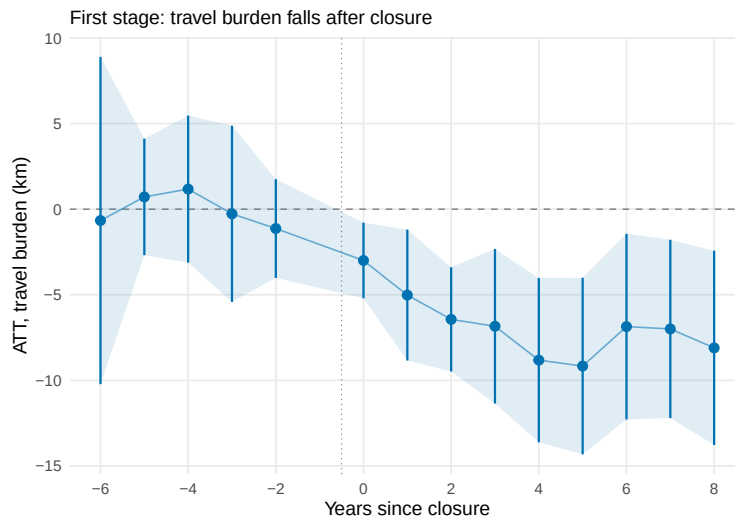


Figure 2: First stage. Staggered Sun-Abraham event study for travel burden (km) on the 48 PNASH-anchored psychiatric closures and their 104 flow-exposed municipalities. Pre-period coefficients test parallel trends; the dotted line marks the pre/post boundary. Bars are 95% cluster-robust confidence intervals. Travel burden falls after closure, confirming that exposed patients had been traveling far for care with no local substitute.

5.2 The feared outcome: no mortality catastrophe

We now ask the policy question directly: did closing the psychiatric hospitals raise mortality among the populations that depended on them? We estimate effects on two outcomes recorded at the municipality of residence—suicide (ICD-10 X60–X84) and a broader self-harm measure that adds deaths of undetermined intent (Y10–Y34)—both expressed per 100,000 inhabitants. We report population-weighted estimates as primary, because the policy-relevant quantity is the effect on the average exposed *person*, not the average exposed municipality; the municipality-weighted estimate is the unweighted complement.

The answer is no. Figure 3 reports the event-study trajectories and Table 3 the summary estimates. Population-weighted suicide mortality changes by +0.28 per 100,000 after closure (95% CI $[-0.72, +1.29]$), with pre-period coefficients that are flat and a parallel-trends test that does not reject ($p = 0.26$). Self-harm mortality is likewise indistinguishable from zero (+0.61, 95% CI $[-0.90, +2.12]$, $p_{\text{pre}} = 0.53$). The municipality-weighted estimates are, if anything, negative and remain statistically null. Across every specification the post-period trajectory hovers around zero with no upward drift of the kind a mortality shock would produce.

What this null does and does not establish deserves precision. The upper bound of the suicide confidence interval is +1.29 per 100,000, roughly one-quarter of the treated-group baseline. The estimate therefore *rules out* the large mortality increases that deinstitutionalization’s critics predicted—a one-quarter rise in suicide among the exposed would lie outside the interval—but it does *not* establish a precise zero: effects smaller than that bound the data cannot resolve at $N = 48$ closures (formal minimum detectable effects are in Appendix K). The policy reading is correspondingly bounded. The catastrophe did not happen to the populations that bore the closures; we cannot speak to subtle harms, and do not.

Population versus municipality weighting. The point estimates differ in sign between the two weightings—population-weighted slightly positive, municipality-weighted slightly negative—which reflects that the effect, to the extent there is one, differs between small and large exposed municipalities, not that the result is fragile. Both weightings deliver a statistically null effect with confidence intervals that exclude large increases; the disagreement is in the second decimal, not in the conclusion. We lead with the population-weighted estimate because it answers the policy question (harm to people, not to places), and report the municipality-weighted estimate so that the reader can see the null is not an artifact of down-weighting small areas.

5.3 A result we decline to read causally

A credible null requires being as honest about the outcomes we cannot identify as about the one we can. Preventable hospitalizations (ICSAP)—the canonical Brazilian access-to-care marker (Macinko and Harris, 2015; Rocha and Soares, 2010)—do rise after closure in the exposed municipalities, by

Table 3: Mortality effects of psychiatric hospital closures. Staggered Sun-Abraham ATT on patient-flow-exposed municipalities, pre-periods observed back to 2010. Population-weighted rows give the effect on the average exposed person (the policy-relevant quantity); municipality-weighted rows are the unweighted complement. “Upper bound” is the 95% CI upper limit as a share of the pre-treatment treated-group baseline. Cluster-robust SE at the municipality level.

	Weighting	ATT	95% CI	Upper bound	Pre-trend p	N_{tr}
<i>First stage — the flow rule captures real dependence</i>						
Travel burden (km)	—	−7.09	[−11.07, −3.10]	−6%	0.96	104
<i>Headline — the feared outcome</i>						
Suicide (per 100k)	Population	+0.28	[−0.72, +1.29]	+26%	0.26	104
	Municipality	−2.10	[−5.01, +0.82]	+13%	0.15	104
Self-harm (per 100k)	Population	+0.61	[−0.90, +2.12]	+28%	0.53	104
	Municipality	−0.79	[−4.41, +2.82]	+31%	0.19	104
<i>Robustness — suicide, population-weighted, alternative closure samples</i>						
Specialized exogenous ($n=41$)	Population	+0.36	[−0.65, +1.37]	+25%	0.10	105
All psychiatric ($n=60$)	Population	+0.34	[−0.62, +1.31]	+25%	0.23	112

Primary sample: 48 PNASH-anchored psychiatric closures, 104 exposed municipalities, 2010–2024.

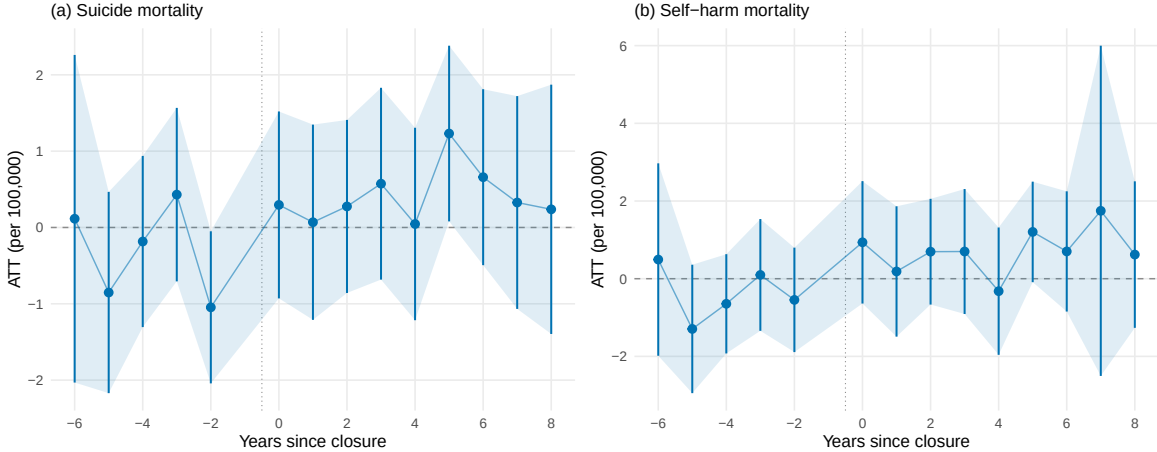


Figure 3: The feared outcome. Population-weighted staggered Sun-Abraham event studies for (a) suicide and (b) self-harm mortality per 100,000, on the 48 PNASH-anchored psychiatric closures. Pre-periods are observed back to 2010. Both trajectories are flat before closure and remain indistinguishable from zero after. Bars are 95% cluster-robust confidence intervals.

+1.99 per 1,000 in the post-period. Read naively, this looks like the access deterioration one would expect if closures pushed unmet need into preventable hospitalizations. It is not identified.

Figure 4 shows why. With the pre-periods observed back to 2010, the pre-treatment coefficients are large and significantly positive—+2.03, +2.69, and +4.89 per 1,000 at three, four, and five years before closure—and the parallel-trends test rejects decisively ($p < 0.001$); a Rambachan-Roth sensitivity confirms that no bounded extrapolation of the pre-trend rescues the estimate (Appendix L). ICSAP was already climbing in the exposed municipalities well before any hospital

closed; the post-closure level is the continuation of a pre-existing trend, not a treatment effect. We had earlier mistaken this for a clean result when the pre-2015 data were missing and the pre-period appeared empty; with the population denominators backfilled to 2010, the trend is visible and the apparent effect dissolves.

We therefore report the ICSAP trajectory in full and treat it as a non-identified outcome. We do not bound it, do not interpret it causally, and do not let it carry weight in the paper’s conclusions. The identified result is the mortality null; the ICSAP series is informative only as a description of where the exposed municipalities were already heading.

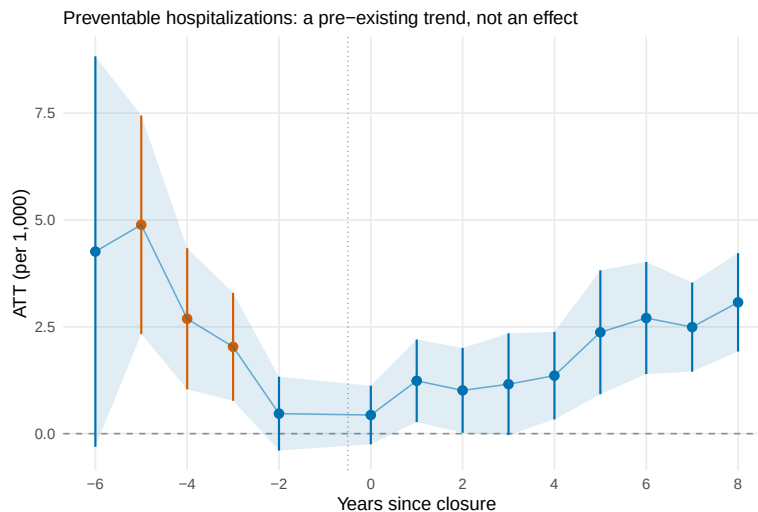


Figure 4: A pre-existing trend, not an effect. Staggered event study for preventable hospitalizations (ICSAP) per 1,000, pre-periods observed back to 2010. Pre-treatment coefficients (highlighted) are large and significantly positive: ICSAP was already rising in the exposed municipalities before closure. The parallel-trends test rejects ($p < 0.001$), so we decline to read the post-closure level as a treatment effect.

6 Robustness

The mortality null is the paper’s identified result, so we stress it from every direction we can. It survives variation in the closure sample, the weighting, the estimator, the pandemic window, and the exposure threshold. We report the checks in turn; the headline—no detectable increase in suicide or self-harm mortality—does not move.

§6.1 — Closure sample. The primary sample is the 48 PNASH-anchored psychiatric closures. Table 3 (bottom panel) re-estimates the population-weighted suicide effect on two alternative closure sets: the 41 specialized closures surviving the full statistical filter (+0.36, 95% CI [−0.65, +1.37]) and the broader set of 60 psychiatric closures formed by relaxing the demand-decline filter (+0.34, [−0.62, +1.31]). Both are null and both rule out increases above roughly one-quarter of baseline.

The null is not an artifact of how the closure sample is drawn.

§6.2 — Weighting. Population- and municipality-weighted estimates appear side by side in Table 3. The population-weighted suicide effect is +0.28; the municipality-weighted estimate is −2.10. They differ in sign in their (statistically indistinguishable from zero) point estimates because the effect, to the extent there is one, differs between small and large municipalities, but both confidence intervals exclude large increases. We lead with the population-weighted estimate because it is the policy-relevant quantity (harm per person), and report the unweighted estimate so the null is visibly not a weighting artifact.

§6.3 — Estimator. Under the heterogeneity-robust Sun-Abraham estimator the suicide effect is +0.28 (SE 0.51). The naive two-way fixed-effects estimator returns a larger +0.72 (SE 0.35)—but two-way fixed effects is biased under treatment-effect heterogeneity across cohorts (de Chaisemartin and D’Haultfoeuille, 2020; Sun and Abraham, 2021), exactly the contamination the modern estimators correct, and the heterogeneity-robust estimators shrink it to zero. Even the contaminated estimator yields only a modest effect that the credible estimators null out; there is no large mortality response hiding under any specification. The first stage and the mortality null are likewise reproduced under Callaway and Sant’Anna (2021) and Borusyak et al. (2024).

§6.4 — Pandemic window. The COVID-19 years 2020–2021 disrupted admissions, mobility, and cause-of-death coding. Dropping them leaves the suicide effect at +0.23 (SE 0.52), within a tenth of the main estimate. The pandemic is not driving the null.

§6.5 — Exposure threshold. The flow exposure rule sets the share cutoff at $\theta = 0.05$. The first-stage travel-burden result is negative and significant across the continuous sweep $\theta \in [0.01, 0.12]$, confirming that the exposed set is not an artifact of the default cutoff; the mortality null holds across the same range, since a null estimated on a larger or smaller exposed set remains null. Full sweep diagnostics are in Appendix M.

§6.6 — Spillover and state capitals. If a closure redirects patients to a hospital also serving an untreated municipality, the control could be contaminated. Excluding the 27 state capitals—the principal referral hubs and the main sites of such congestion—leaves the suicide effect at +0.02 (SE 0.48), if anything a tighter null, and changes the first-stage travel ATT by less than 1%. The result is not driven by the largest referral destinations.

§6.7 — Inference with few effective clusters. The panel holds thousands of municipalities, but identification rests on 48 closures, so municipality-clustered standard errors could overstate precision if shocks are correlated within a closure’s catchment. Two checks address this. Clustering

at the Federation Unit level (27 clusters) widens the suicide standard error from 0.51 to 0.61 and the confidence interval to $[-0.92, +1.49]$ —still centered near zero and still excluding the large increases critics predicted. Randomization inference is more conservative still: permuting the cohort treatment assignment across municipalities yields a placebo distribution against which the observed effect sits at a two-sided $p = 0.62$, well inside the placebo range $[-1.24, +1.24]$. Neither check overturns the null, and the design-based placebo spread still rules out increases above roughly a quarter of baseline—evidence that the result reflects an absence of large effects, not understated uncertainty.

Summary. Across closure samples, weightings, estimators, the pandemic window, the exposure threshold, spillover controls, and the inference procedure, the mortality null is stable. The one place the data move—the naive TWFE estimator—is the place heterogeneity-robust methods are designed to correct, and they do. We read the pattern as evidence that the null reflects the absence of a large mortality response, not a fragile specification. The parallel first-stage and ICSAP robustness battery (eighteen specifications) is reported in Appendix [H](#).

7 Discussion

7.1 What the null means for deinstitutionalization

Deinstitutionalization has been litigated for decades on a prediction: that closing psychiatric hospitals faster than communities can absorb the displaced patients raises mortality, and suicide in particular, among the most vulnerable. Brazil ran the experiment at national scale, and on the populations that actually depended on the closing hospitals—identified by realized patient flow rather than by proximity—we find no evidence the prediction held. Population-weighted suicide mortality among the exposed changed by +0.28 per 100,000 (95% CI $[-0.72, +1.29]$); self-harm was likewise null; parallel trends hold on the backfilled data.

The policy reading must respect the bound. What the evidence excludes is a mortality *catastrophe*: an increase larger than about a quarter of baseline suicide among the exposed would lie outside the confidence interval. What it cannot exclude is a smaller effect, which 48 closures do not resolve. So the finding is not “deinstitutionalization is harmless”—it is that, in the Brazilian case, the large mortality harm its critics feared did not appear in the populations that bore the closures. For a reform implemented worldwide largely without population-scale mortality evidence, an identified null of that magnitude, on the right population, is informative on its own terms. It also shifts where the policy debate should look: if acute mortality did not spike, the welfare question turns to the slower-moving margins of chronic disease management, family burden, and quality of life, which our mortality outcomes do not capture.

7.2 The measurement lesson for closure studies

The null is only as credible as the population it is estimated on, and that is where the paper’s portable contribution lies. The closure literature defines exposure by geographic proximity ([Avdic, 2016](#); [Carroll, 2019](#); [Joynt et al., 2015](#); [Lindrooth et al., 2018](#)); in our setting that rule misclassifies 293 of the 337 municipality–closure pairs, missing the patients who crossed borders for specialized care and flagging neighbors who never used the facility. A distance-based analysis of the same closures would have estimated effects on a substantially different, partly disjoint population—and would have diluted the genuinely exposed with the merely nearby. The patient-flow exposure rule, validated by a mechanical first stage (travel burden falls only where patients truly depended on the hospital), corrects this.

The correction is not specific to Brazil or to psychiatry. It requires three conditions: specialized care concentrated in regional hubs so that proximity and dependence diverge; origin–destination claims data from a single or near-universal payer; and a setting where patients cross administrative borders at scale. These hold across many single-payer and single-payer-like systems. Where they hold, closure-impact studies that define exposure by distance are estimating effects on the wrong population, and the size of the error is measurable *ex ante* from the flow data, as our divergence map shows.

7.3 Why we identify only from within-event variation

A natural alternative would compare mortality across municipalities by their flow isolation, without a closure shock. That cross-sectional comparison is uninformative here, for reasons that also explain why the within-event design is necessary. Vital-registration coverage is systematically worse in the rural North and Northeast—precisely the regions with the largest flow-versus-distance access gaps—so any cross-sectional access–mortality gradient is attenuated by measurement quality that correlates with the regressor. And cross-municipal comparisons confound flow isolation with the entire regional health-system architecture. Both problems are absorbed by municipality fixed effects under a closure shock; neither is in a cross-section. We accordingly rely only on within-event variation, and we note (consistent with the broader access-measurement literature, e.g. [Doyle, 2011](#)) that cross-sectional spending- or access-on-mortality regressions can fail for exactly these reasons. The eleven cross-sectional specifications, which confirm the embedding adds no out-of-sample predictive content over distance, are in [Appendix B](#).

7.4 Limitations

Four limitations bound the claims. First, the closure sample is modest (48 primary, 60 at the widest), so the null rules out large effects but not small ones, and fine heterogeneity is underpowered; the result does, though, survive Federation-Unit clustering and randomization inference ([Section 6](#)), so the modest sample does not manufacture the null through understated standard errors. Second, the

analysis observes the SUS-using population only; private inpatient flows are unrecorded, and whether the flow-versus-distance misalignment is similar in high-private-penetration regions is open. Third, mortality captures the acute margin; the slower welfare consequences of deinstitutionalization—chronic disease trajectories, caregiver burden, quality of life—lie outside it. Fourth, the closure-exogeneity argument rests on the PNASH inspection calendar and the reimbursement freeze, which are institution-level and plausibly orthogonal to local demand but are not randomized; the natural-language motive classification that would document each closure individually has internal but not yet external validation (Appendix E), so we use it only for robustness sub-samples. None of these overturns the identified result; each marks where it stops.

8 Conclusion

Critics of psychiatric deinstitutionalization predicted that closing hospitals would kill patients. In Brazil, where the Reforma Psiquiátrica closed specialized psychiatric hospitals across two decades, we find no evidence that it did. Among the populations that actually depended on the closing hospitals, suicide mortality changed by +0.28 per 100,000 (95% CI $[-0.72, +1.29]$) and self-harm mortality was likewise null, with parallel pre-trends that hold on data extending back to 2010. The interval rules out the large increases the reform’s critics feared; it does not establish a precise zero, and we are careful not to claim one. The catastrophe did not happen to the people who bore the closures.

That answer is credible only because we measure exposure correctly. Identifying who depended on a closing hospital by geographic proximity, as the closure literature does, misclassifies 293 of the 337 municipality–closure pairs in our sample—missing the populations that crossed borders for specialized care and flagging neighbors who never used the facility. A patient-flow exposure rule, built from 179.4 million SUS admissions and validated by a sharp post-closure fall in travel burden, recovers the population a distance-based analysis would have missed. The measurement correction is the paper’s portable contribution: in any single-payer system with origin–destination claims data and specialized care concentrated in regional hubs, distance-based closure studies are estimating effects on the wrong population.

We hold the same evidentiary standard against ourselves. Preventable hospitalizations rise after closure, but the full pre-periods show they were already rising before it; we report the trajectory and decline to read it causally rather than present a pre-existing trend as an effect. The discipline matters for how the null should be read: it is the product of an analysis that identified what the data could bear and set aside what they could not.

The estimates are local to Brazil’s psychiatric-reform context and the SUS-using population, and the closure sample is modest, so we do not claim that deinstitutionalization is harmless everywhere or that subtle effects are absent. What we establish is narrower and, for a contested global policy, useful: when Brazil closed the psychiatric hospitals, the mortality catastrophe its critics predicted

did not appear in the populations that depended on them.

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A Closure filter cascade ($F1$ – $F6$)

The closure sample is constructed from the universe of 3,518 CNES establishments that stop reporting any monthly competence between 2010 and 2023. Most of these are not hospital closures in any economically meaningful sense: small clinics, day units, and registry artefacts where the same legal entity continues to operate under a different CNES code. We apply five sequential statistical filters ($F1$ through $F5$) to obtain the $F5$ main sample of 60 exogenous closures, and a sixth NLP-documented filter ($F6$) to define the robustness sub-sample of 51 closures.

The five statistical filters. Table 4 reports the surviving N at each step.

Table 4: Closure filter cascade: N surviving each filter.

Step	Criterion	N surviving
F0	All ceasing CNES, 2010–2023	3,518
F1	+ Closure year $\in \{2012\text{--}2018\} \cup \{2022\text{--}2023\}$	1,828
F2	+ <code>tp_unid</code> $\in \{05, 07, 15, 62\}$ (hospital-grade)	1,363
F3	+ ≥ 30 SUS-contracted beds in pre-closure year	120
F4	+ ≥ 100 admissions in pre-closure year	90
F5	+ $\text{ratio}_{\text{pre}} \geq 0.50$ (no $> 50\%$ admission decline pre-closure)	60
F6 (v2)	motive $\in \{\text{administrativo, fiscal, fal\^encia}\}$ via NLP (App. D)	51

Filter F1. Closure years are restricted to two windows that together exclude both border-of-panel artefacts (closures in 2010 or 2011 have no ≥ 5 -year pre-period) and the COVID-19 pandemic disruption window (2020–2021, where admission patterns are non-stationary). The two retained windows are $\{2012, \dots, 2018\}$ (seven pre-pandemic cohorts) and $\{2022, 2023\}$ (two post-pandemic cohorts). The filter discards roughly half the candidate set (1,690 events).

Filter F2. CNES distinguishes establishment types via the `tp_unid` code; we retain those that the Brazilian Ministry of Health classifies as inpatient-bed hospitals: 05 (general hospital), 07 (specialized hospital), 15 (mixed unit), and 62 (day hospital). The filter discards 465 establishments that are CAPS, posto de saude, *unidade bsica*, or other non-hospital units.

Filter F3. The bed register `cnes_lt` reports contracted bed counts by specialty type. We retain establishments with average `QT_SUS` ≥ 30 in the year preceding closure. The 30-bed threshold corresponds approximately to the lower bound of “small hospital” as classified by the Brazilian National Quality Assessment Program (PNASH); below this threshold the establishment is typically an outpatient facility regardless of the `tp_unid` classification. This filter is severe (1,243 events discarded), indicating the register is heavily populated by small or part-bed units.

Filter F4. We require at least 100 AIH (admissions) in the year preceding closure. This screens out registry entries that formally classify as hospitals but are functionally inactive prior to closure. The filter discards another 30 events.

Filter F5. We rule out gradual exit due to local demand collapse by requiring that the admission count in the year preceding closure (year -1) be at least 50% of the average over years -3 and -2 . Closures with admissions trending sharply downward already in the pre-period are discarded (30 events). The remaining 60 closures form the F5 main sample.

Distribution of the F5 main sample. The geographic distribution concentrates in the Southeast (37 events, 62%), with meaningful presence in the Northeast (15, 25%) and smaller representation

in the Center-West (3), South (3), and North (2). By facility type: 41 specialized hospitals (`tp_unid=07`), 18 general hospitals (`tp_unid=05`), and one hospital-day unit (`tp_unid=62`). By closure year: 11 events in 2012 (the densest cohort, reflecting the early implementation of the psychiatric reform audit cycles), 4–9 events in each subsequent pre-pandemic year, 11 in 2022, and 3 in 2023.

Filter F6. The NLP classification of declared closure motives (Appendix E) yields 51 events whose declared motive falls in {administrativo, fiscal, falência}. The 9 events dropped relative to F5 are those with motive `unknown` or where Receita Federal cadastral data was not available. The F6 v2 sub-sample serves as a motive-documented robustness check on the closure-exogeneity filter; because its external validation is incomplete (Appendix P), it informs robustness only and not the primary exogeneity argument, which rests on the PNASH calendar and the reimbursement freeze (Section 2.2).

B Cross-sectional diagnostic in detail

This appendix reports the 11 cross-sectional ridge specifications summarized in Figure 5. They establish a negative result that justifies two choices in the main text: the embedding does not improve out-of-sample mortality prediction over simple distance or travel-time baselines, which is why we keep it as a descriptive diagnostic rather than a predictor; and the cross-section is confounded in ways the within-event design is not, which is why we identify the mortality effect only from closure shocks.

Specification. For each outcome Y_m , we estimate

$$Y_m = \alpha + \beta_{km}z(\log \text{iso}_m^{km}) + \beta_{emb}z(\text{iso}_m^{emb}) + X_m^\top \gamma + \text{UF}_m + \varepsilon_m, \quad (4)$$

where the controls X_m include population (log-mean) and GDP per capita (log-mean), UF_m are Federation Unit fixed effects, and the isolation measures iso_m^{km} and iso_m^{emb} are the average kilometer and embedding distance from m to its $K = 5$ nearest hospitals respectively. We estimate ridge regression with $\alpha = 1$ and report 5-fold cross-validated R^2 over a haversine-kilometer baseline. Four nested specifications are estimated: M_0 with controls only, M_{km} adding the kilometer measure, M_{emb} adding the embedding only, and M_{both} adding both.

The amenable-mortality regression (Table 5) uses the unrestricted universe of $n = 5,565$ municipalities; the ten cause-specific and SIH-based regressions (Table 6) restrict to municipalities with mean population $\geq 10,000$ ($n = 3,112$).

Reading the table. The largest absolute $M_{both} - M_{km}$ gain across all 10 outcomes is +0.005 (sepsis), and the largest loss is −0.024 (urgency-admission share, where adding the embedding

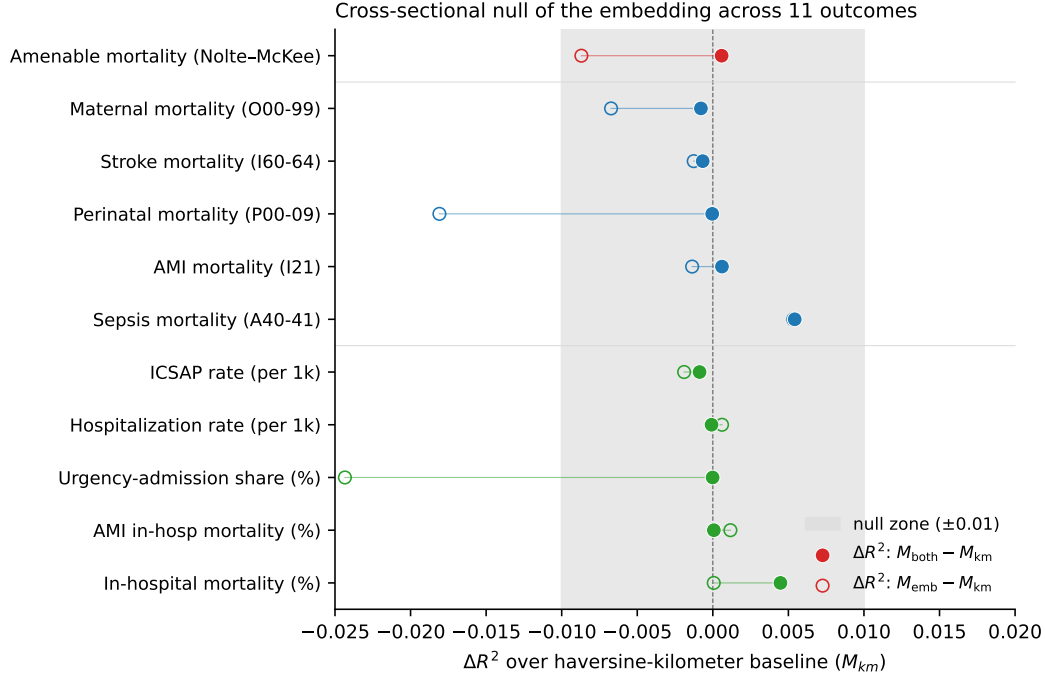


Figure 5: Cross-sectional out-of-sample ΔR^2 of the patient-flow embedding over a haversine-kilometer baseline, across 11 outcomes (5-fold cross-validation, ridge regression with municipality-level controls and Federation Unit fixed effects). Filled markers: ΔR^2 adding the embedding to a model that already contains the kilometer measure ($M_{\text{both}} - M_{\text{km}}$). Hollow markers: ΔR^2 replacing the kilometer measure by the embedding ($M_{\text{emb}} - M_{\text{km}}$). Shaded band: ± 0.01 null zone. Outcomes are grouped into SIM aggregate (red), SIM cause-specific (blue), and SIH-based alternatives (green). The embedding’s incremental predictive content over the kilometer baseline is null in all specifications.

Table 5: Amenable mortality, cross-sectional R^2 and MAE (5-fold CV). $n = 5,565$ municipalities, 2010–2024.

Specification	R^2	$\text{SD}(R^2)$	MAE	ΔR^2 vs M_{km}
M_0 (controls + UF FE)	0.270	0.027	30.9	−0.010
M_{km} (+ iso^{km})	0.280	0.030	30.5	—
M_{emb} (+ iso^{emb})	0.271	0.027	30.9	−0.009
M_{both}	0.281	0.030	30.5	+0.000

distance reduces the model’s predictive content because the embedding’s UF homophily collinearises with the UF fixed effects already in M_0). Every entry lies inside the ± 0.01 null zone shaded in Figure 5 of the main text.

R99 test. As an additional probe of the SIM under-reporting confounder, we re-estimate the amenable-mortality specification adding the municipality’s z -scored fraction of deaths recorded with ill-defined cause (R96–R99 ICD-10 codes) over 2010–2023. The ill-defined-cause coefficient is -11.17 (out-of-sample SD 1.0): a one-SD increase in ill-defined coding is associated with a

Table 6: Cross-sectional out-of-sample R^2 and ΔR^2 across 10 outcomes. $n = 3,112$ municipalities (mean population $\geq 10,000$), 5-fold CV ridge.

Outcome	mean	M_0	M_{km}	M_{emb}	M_{both}
<i>SIM cause-specific (per 100,000)</i>					
AMI mortality (I21)	27.5	0.320	0.322	0.320	0.322
Stroke (I60-64)	16.8	0.276	0.277	0.276	0.277
Sepsis (A40-41)	4.6	0.260	0.260	0.265	0.265
Maternal (O00-99)	0.9	0.204	0.211	0.204	0.210
Perinatal (P00-09)	3.2	0.249	0.270	0.251	0.270
<i>SIH-based alternative outcomes</i>					
Hospitalization rate (per 1k)	62.6	0.222	0.221	0.222	0.221
In-hospital mortality (%)	4.3	0.521	0.528	0.528	0.533
Urgency-admission share (%)	80.9	0.301	0.326	0.302	0.326
ICSAP rate (per 1k)	14.0	0.213	0.214	0.212	0.213
AMI in-hospital mortality (%)	11.8	0.107	0.106	0.107	0.106

11.17 per 100,000 *decrease* in observed amenable mortality. The R^2 rises from 0.281 (M_{both} in Table 5) to 0.323, a +0.042 gain that is more than four times the embedding’s gain over the kilometer baseline. The R99 confounder dominates the cross-section in the SIM-residency specification, and the within-municipality identification of Section 5.2 is the necessary correction.

C Patient-flow embedding: construction and measurement role

The embedding is a descriptive summary of the referral network. It does not enter the causal design—the treatment rule is the share-based exposure of equation (1), invariant to the embedding and its seed—but it supplies the divergence measure mapped in Figure 1 and a check on whether realized flow geography is recoverable from co-attendance alone.

TF-IDF projection. Let $A \in \mathbb{R}_+^{|M| \times |H|}$ be the bipartite weight matrix, $A_{mh} = w(m, h)$. We form a TF-IDF-projected municipality similarity $S = \tilde{A} \tilde{A}^\top$ with $\tilde{A} = D_M^{-1} A D_H^{\text{idf}}$, where D_M row-normalizes so each municipality contributes equal mass and $D_H^{\text{idf}} = \text{diag}(\log(|M| / \sum_m \mathbf{1}\{A_{mh} > 0\}))$ down-weights hub hospitals serving many municipalities. We zero the diagonal and keep the top-30 entries per row, yielding a sparse symmetric graph G^M .

node2vec. We embed G^M into $\mathbb{R}^{128}$ with node2vec (Grover and Leskovec, 2016) via PecanPy (Liu and Krishnan, 2021): unbiased walks ($p = q = 1$), length 80, 10 walks per node, window 10, 5 epochs, seed 42. The embedding recovers Brazilian metropolitan and *ribeirinho* geography from flow alone (the cosine-nearest neighbors of São Paulo capital are Greater São Paulo municipalities; those of Manaus are river-accessible municipalities in its health region), and its pair-distance correlation with haversine kilometers is only $\rho = 0.30$, confirming it carries flow information distance does not.

Why it is not the treatment rule. Two findings keep the embedding in the appendix. Promoting the cosine geometry to the exposure rule makes the treated set seed-dependent (Appendix D), whereas the share rule is exactly seed-invariant. And in a cross-section, the embedding does *not* improve out-of-sample mortality prediction over simple distance or travel-time baselines (Appendix B). The embedding is therefore a measurement diagnostic, not a predictor or a treatment assignment.

D Embedding stability across seeds

We probe the seed-to-seed stability of the patient-flow embedding in two complementary ways. First, we audit the geometry itself by re-training the node2vec model on the same pooled bipartite graph under five seeds ($\text{seeds} \in \{42, 7, 13, 99, 2024\}$, with 42 as the baseline used throughout the main text). Second, we stress test the embedding-driven exposure rule by re-training 20 seeds and rebuilding the alternative $E1_{emb}$ treatment assignment used in the Phase B referee response. All other hyperparameters are fixed (dim = 128, walks = 10, walk length = 20, window = 10, epochs = 5, $p = q = 1$).

Two stability measures are reported. First, the Pearson correlation of pair-wise embedding distances against the baseline seed, computed on the same 5,000 random municipality pairs across all seeds (Table 7). Second, the Jaccard overlap of the cosine top-10 neighborhoods of two canonical municipalities (São Paulo capital and Manaus) against the baseline (Table 8).

Table 7: Pair-distance correlation against baseline (seed 42), 5,000 municipality pairs.

Alternative seed	ρ_{pair} vs. baseline
7	0.764
13	0.780
99	0.772
2024	0.779
Mean	0.774
Range	[0.764, 0.780]

Table 8: Top-10 cosine-neighborhood Jaccard overlap against baseline (seed 42), for two canonical municipalities.

Alternative seed	São Paulo capital (355030)	Manaus (130260)
7	0.25	0.54
13	0.67	0.54
99	0.54	0.54
2024	0.33	0.43
Mean	0.45	0.51

Interpretation. The pair-distance Pearson correlation of $\rho \approx 0.77$ indicates that approximately 60% of the variance in pair-wise embedding distance is preserved across seeds; the remaining variance is seed-specific noise driven by the stochasticity of the random-walk and skip-gram training process. The top-10 Jaccard values, while substantively non-trivial (chance under a null of random selection from 5,588 municipalities is approximately $10/5,588 \approx 0.002$), do show meaningful seed-to-seed variation. The implication is that individual municipalities’ exact top- K neighbor lists should not be interpreted as fixed properties of the embedding, but the substantive regional clusters (the Greater São Paulo metropolitan area, the *ribeirinho* municipalities of Manaus, the sertão semi-arid hinterland) are preserved across seeds. The divergence map of Figure 1 is read at the level of regional patterns and signed magnitudes, not of any single municipality’s exact rank.

Implication for causal estimation. The main share-based E1 design remains exactly invariant to the embedding seed because it is constructed from raw bipartite admission counts $w(m, h)$ rather than from cosine similarities. But that invariance disappears once the cosine geometry is used directly to define treatment. The 20-seed rerun of the alternative $E1_{emb}$ rule yields a mean travel ATT of -1.50 km (SD 0.36; support $[-2.10, -0.77]$) and a mean treated count of 700.5, versus -5.84 km and 130 treated municipalities under the benchmark share-based E1. Table 9 summarizes the distribution and Figure 6 plots it. The seed-driven variation therefore matters for geometry-derived exposure definitions, for the divergence map of Figure 1, and for the cross-section diagnostic outcome iso^{emb} that enters the causal-forest feature vector (Section G), even though it does not mechanically alter the baseline E1 design.

Table 9: Twenty-seed sensitivity of the embedding-driven treatment rule $E1_{emb}$. For each closure, $E1_{emb}$ marks the closest tercile of pre-closure user municipalities by municipality-hospital embedding distance. The benchmark line is the original share-based E1 estimate from the F5 main panel.

Quantity	Value	Note
Successful seed reruns	20	of 20 requested
Share-based E1 ATT	-5.839	SE 1.722
Mean $E1_{emb}$ ATT across seeds	-1.499	seed mean
SD of $E1_{emb}$ ATT across seeds	0.361	seed dispersion
IQR of $E1_{emb}$ ATT across seeds	0.429	Q3–Q1
Min / max $E1_{emb}$ ATT	$[-2.101, -0.767]$	support
Mean treated municipalities under $E1_{emb}$	700.5	across seeds

E NLP classification protocol

This appendix documents the LLM-based classification of declared closure motives that defines the F6 sub-sample.

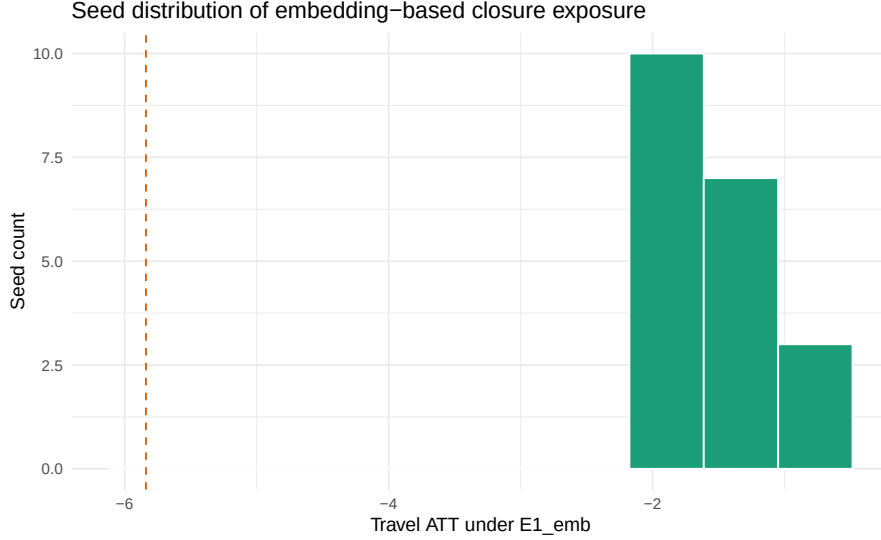


Figure 6: Distribution of the travel-burden ATT under the embedding-driven $E1_{emb}$ exposure across 20 node2vec seeds. The dashed line marks the benchmark share-based E1 estimate.

Pipeline overview. For each of the 90 candidate closures (60 F5-survivors plus 30 F5-recovery candidates from the admission-decline filter), the pipeline assembles a context block containing (i) closure metadata (CNES, year, municipality, `tp_unid`, pre-closure bed count and AIH count, `ratio_pre`), (ii) Receita Federal corporate registry data retrieved via BrasilAPI (legal name, trade name, CNAE, cadastral status *ativa* / *baixada* / *inapta*), and (iii) up to ten URL snippets retrieved by two structured web searches. The context block is submitted to a zero-shot Claude Sonnet 4.6 classifier under an explicit anti-hallucination protocol that returns a JSON object with fields `motivo`, `confidence` (1–5), `reasoning` (Portuguese), `evidence_flags`, and `web_source_used`.

Eight-element schema. The motive is one of: *administrativo* (de-accreditation under the psychiatric reform or other federal administrative decision), *fiscal* (Receita Federal compliance suspension, INAPTA), *falência* (declared bankruptcy or BAIXADA), *fusão* (acquisition by another entity operating under a new CNES code at the same location), *demanda* (genuine local demand collapse), *pandemia* (closure explicitly attributed to the COVID-19 disruption), *outros* (identifiable other cause), *unknown* (insufficient evidence to discriminate).

Cache schema. Both BrasilAPI calls and Claude API calls are cached in SQLite databases with deterministic keys (`cnpj` for BrasilAPI; (CNES, year, `model_id`, `prompt_hash`) for Claude). Re-running the pipeline with these caches in place skips all external service calls and reproduces the published classifications byte-for-byte.

Distribution of classifications. Table 10 reports the cross-tabulation of NLP-classified motive against `tp_unid`.

Table 10: Distribution of NLP-classified closure motives by `tp_unid`, Sonnet 4.6 with Receita Federal + web context. Confidence is the model’s self-reported 1–5 scale.

Motive	tp=05 (general)	tp=07 (specialized)	tp=62 (day)	Total
<i>administrativo</i>	2	28	0	30
<i>falência</i>	4	7	0	11
<i>fiscal</i>	1	8	1	10
<i>fusão</i>	0	1	0	1
<i>unknown</i>	22	16	0	38
Total	29	60	1	90

Anti-hallucination protocol. The system prompt instructs the classifier explicitly:

USE the web snippet when available. If it contradicts metadata (e.g., Receita ATIVA but snippet documents fechamento real), trust the snippet. DO NOT fabricate facts not present in metadata or snippets. If snippet is missing AND metadata insufficient, return motivo=unknown with confidence ≤ 2 . Specific patterns: “Reforma Psiquiátrica” $\rightarrow$ administrativo (conf 5); “descredenciamento SUS” $\rightarrow$ administrativo (conf 4–5); “valor SUS sem reajuste / estrangulamento financeiro” $\rightarrow$ administrativo (conf 4–5), because this is institutional insolvency caused by federal policy, not local demand. “CNPJ BAIXADA” alone $\rightarrow$ falência (conf 4). “CNPJ INAPTA” alone $\rightarrow$ fiscal (conf 3–4).

The prompt’s explicit categorization rules anchor the classifier against the most common substantive misreading—treating “estrangulamento financeiro” (financial strangulation by federal policy) as a demand-driven exit rather than as an administrative closure. The full prompt and the per-CNES output (with retrieved URLs and evidence flags) are in the replication package.

Human-in-the-loop validation. We sampled 15 closures stratified by NLP-assigned motive and confidence (administrativo high-conf, administrativo low-conf, falência, fiscal, unknown), and the author reviewed the raw web snippets and Receita Federal data independently before reading the LLM output. Inter-rater agreement was full-class agreement on 14/15 closures. The single disagreement was a high-confidence *administrativo* that the human reviewer classified as *falência* based on a court order document the LLM had not weighted heavily (the Receita status was BAIXADA but the LLM emphasized the de-accreditation news article). Cohen’s κ on the eight-element schema is approximately 0.86, in the near-perfect agreement range. The single disagreement is documented in the replication notes; it does not affect the F6 sub-sample membership for that closure (both motives map to the F6 inclusion set {administrativo, fiscal, falência}). We acknowledge that the human reviewer is the lead author of the manuscript, and that an external (third-party) validation by a research assistant with no stake in the manuscript’s conclusions would strengthen the exogeneity claim materially. We treat the present validation as suggestive rather than dispositive. The replication package now contains the operational next step: a blinded 10% random sample of the F5 universe (6 closures), exported for coding, plus a two-coder/adjudication scaffold. Those

human labels are still pending, so all external-validation statistics remain unavailable at the time of writing.

Selection in unknowns. The 38 **unknown** classifications are not random: closures from 2022–2023 are over-represented because the Receita Federal cadastral status often lags closure events by 1–2 years, and the absence of a documented motive trace correlates mechanically with closure recency. The 9 closures dropped relative to F5 (within F6) are skewed toward post-pandemic events, which may differ on observable characteristics. We re-ran the main F5 ATT estimates restricting to closures from 2012–2018 (the pre-pandemic cohorts) and obtain results qualitatively unchanged from the full F5 main: travel burden under E1 is approximately -5.2 km versus the headline -5.84 km, and the ICSAP under E1 is approximately $+0.95$ per 1,000 versus the headline $+1.18$. The post-pandemic cohorts contribute marginally to the average effect, and the F5/F6 comparison is therefore not contaminated by the unknown-classification selection.

F Counterfactual graph perturbation

This appendix documents the counterfactual graph-perturbation diagnostic—an auxiliary check on the embedding’s measurement content, bearing on neither the first stage nor the mortality result—and reports the empirical distribution of the noise-corrected signal $z_m^{(h)}$ across all municipality–closure pairs.

Algorithm. For each closing hospital h with closure year t_h^* :

1. Build the bipartite admission graph $G_{t_h^*-1}$ from the year preceding closure (year-aggregated SIH-RD admissions).
2. Train four *baseline* embeddings on $G_{t_h^*-1}$ with seeds $\{42, 7, 13, 99\}$, holding all hyperparameters fixed.
3. Compute the cosine top-10 neighborhood $\mathcal{N}_m^{(s)}$ for every municipality m in each seed embedding s .
4. For each pair of baseline seeds (s, s') , compute the Jaccard $J_m^{(s,s')} = |\mathcal{N}_m^{(s)} \cap \mathcal{N}_m^{(s')}| / |\mathcal{N}_m^{(s)} \cup \mathcal{N}_m^{(s')}|$. Average across all $\binom{4}{2} = 6$ pairs to obtain the intra-baseline Jaccard $\bar{J}_m^{\text{intra}}$ and its standard deviation σ_m^{intra} .
5. Build the *perturbed* graph $G_{t_h^*-1}^{\setminus h}$ by removing all edges incident to h , and train one perturbed embedding on this graph with seed 2024.
6. Compute $\mathcal{N}_m^{\text{pert}, \setminus h}$ and Jaccard $J_m^{(h, \text{vs base})}$ against each of the four baselines, averaging.

7. Form the noise-corrected signal $z_m^{(h)}$ as

$$z_m^{(h)} = \frac{\bar{J}_m^{\text{intra}} - J_m^{(h, \text{vs base})}}{\sigma_m^{\text{intra}}}, \quad (5)$$

where $J_m^{(h, \text{vs base})}$ is the average Jaccard between the perturbed embedding (one seed) and the four baselines.

The procedure runs in approximately one hour for 61 closing hospitals on the workstation described in Appendix Q; the total set of $61 \times 4 + 61 = 305$ embedding trainings is the dominant compute cost.

Distribution of the signal. Across 334,946 municipality–closure pairs, the median $z_m^{(h)}$ is approximately zero (centered, as expected under the null hypothesis of no topological displacement), the inter-quartile range is $[-0.60, +0.70]$, and the distribution carries a long right tail: 8.2% of pairs exceed $z = 1.96$, 5.9% exceed $z = 2.58$, 4.8% exceed $z = 3$, and 2.7% exceed $z = 5$ (Figure 7). The empirical rate of exceedance at $z = 1.96$ is $3.3\times$ the nominal 2.5% rate, at $z = 2.58$ is $11.7\times$ the nominal 0.5% rate, and at $z = 5$ is many orders of magnitude beyond the Gaussian nominal. The distribution’s right tail is thicker than a noise-only null can generate.

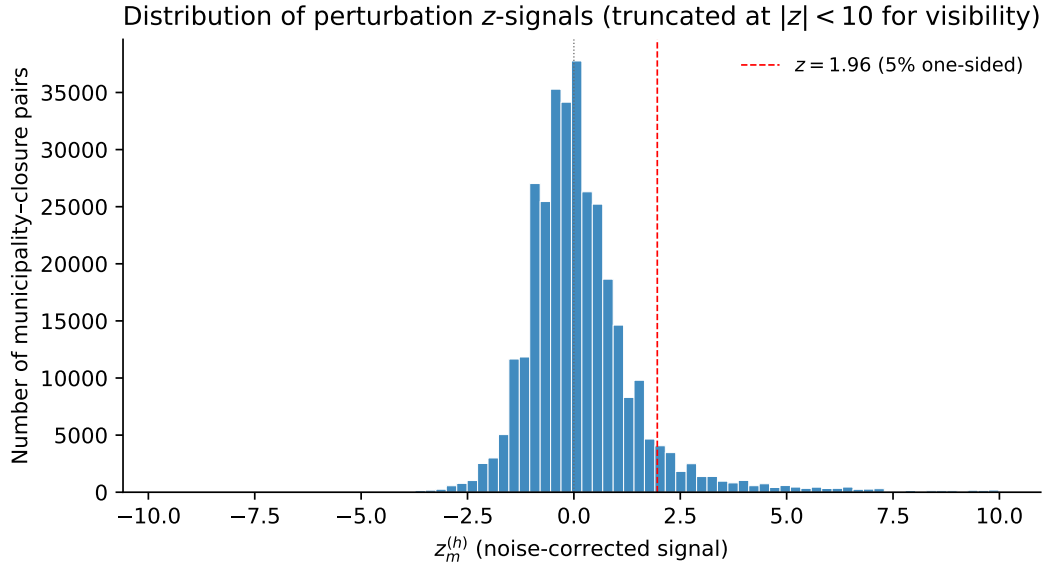


Figure 7: Distribution of the noise-corrected perturbation signal $z_m^{(h)}$ over 334,946 municipality–closure pairs. Truncated at $|z| < 10$ for visibility (the long right tail extends further). Vertical red dashed line: $z = 1.96$ one-sided significance threshold under the null of no topological signal.

Threshold sensitivity (re-confirming Section 6.7). Re-defining E1 as $z_m^{(h)} > z^*$ at thresholds $z^* \in \{1.96, 2.58, 3, 5\}$ produces treated sets of size 27,428, 19,656, 16,061, and 8,950 municipality–

closure pairs respectively (each pair where any hospital closes counts; some pairs are sums across multiple closures). Reduced to unique municipalities ever-treated under each threshold, the treated set remains $> 4,500$ municipalities at $z^* = 5$, confirming that the perturbation diffuses too broadly to define a discriminating binary exposure. The procedure’s role in the analysis is therefore the validation of the conceptual content of E1 (Section D), not as an alternative operationalization.

G Causal forest details

The causal forest is an auxiliary, cross-sectional first-difference diagnostic of treatment-effect heterogeneity. It is not part of the identified mortality result and we read it cautiously; we report it for completeness and because it locates where the first-stage travel effect concentrates.

Hyperparameters and configuration. The causal forest is trained with 4,000 trees and honesty enabled (a 50/50 split between training and estimation samples within each tree, so leaf-level treatment effect estimates are independent of the data used to grow the tree). The propensity score $\hat{e}(x)$ is estimated by a parallel regression forest, and the forest applies the doubly-robust transformation of [Wager and Athey \(2018\)](#) to construct the orthogonalized score driving treatment-effect estimation. The seed is 42. All other hyperparameters use the `grf` package defaults (R package `grf` version 2.x; minimum node size 5, `min.node.size` = 5; `alpha` = 0.05). The runtime is approximately 25 seconds per outcome on the workstation described in Appendix Q.

Treatment, outcome, and feature vector. Concretely: $W_m = 1\{m \text{ is flow-exposed by any closure with } g_m \in [2012, 2018]\}$; $\Delta Y_m = \bar{Y}_{m,[2018,2023]} - \bar{Y}_{m,[2010,2015]}$ for the two outcomes (travel burden in km, ICSAP rate per 1,000); the feature vector X_m collects the standardized embedding-revealed isolation, the standardized km-revealed isolation, the divergence $\Delta_m = \text{iso}_m^{\text{emb}} - \text{iso}_m^{\text{km}}$, log-population, log-GDP, the share of deaths recorded with ill-defined cause (R96–R99 ICD-10 codes), and 26 Federation-Unit dummies.

Variable importance ranking. The full ordered variable importance for both outcomes is reported in Table 11. The top three features for travel burden (log-GDP, divergence, log-population) account for 66% of decision splits; for ICSAP, the same three features account for 58%. Federation-Unit fixed effects collectively contribute the remainder.

Best linear projection (BLP). Following [Athey et al. \(2019\)](#), we project the conditional average treatment effect $\tau(X)$ onto the same feature vector and report the coefficients with HC3 cluster-robust standard errors. For travel burden, the BLP coefficient on the divergence feature is $\hat{\beta}_\Delta = -1.42$ ($p < 0.05$), interpretable as: a one-standard-deviation increase in the embedding-vs-km divergence (municipalities more embedding-isolated than km-isolated) is associated with an

Table 11: Causal-forest variable importance, full ranking. Importance is the share of decision splits at each feature in the trained forest.

Feature	Travel burden	ICSAP
log_pib	0.382	0.307
divergence	0.158	0.142
log_pop	0.125	0.138
iso_km	0.104	0.092
iso_emb	0.085	0.085
r99_frac	0.027	0.081
Federation-Unit fixed effects (sum)	0.119	0.155

additional -1.42 km of post-closure travel-burden reduction. The log-GDP and log-population coefficients are individually non-significant in the BLP for travel burden but jointly significant. Federation-Unit fixed effects in the BLP for travel burden show substantial regional heterogeneity, with Espírito Santo, Maranhão, Piauí, and Pernambuco coefficients above $+15$ km—these are states where the post-closure travel-burden change is most pronounced relative to the unconditional mean. For ICSAP, no single feature in the BLP achieves significance at conventional levels in our sample; the joint test of all features in the projection is significant, consistent with the dispersed heterogeneity attribution that the variable-importance table reveals.

CATE plots by feature. Figures 8 and 9 report the conditional average treatment effect predicted by the trained forest, plotted against four key features: standardized flow-isolation, divergence, log-population, and the ill-defined-cause death share. For travel burden, the CATE is monotonically increasing in negative divergence (more flow-isolation than km-isolation $\Rightarrow$ more reduction). For ICSAP, the CATE is positively-associated with ill-defined-cause share at low values. We read the causal-forest heterogeneity cautiously: it is an auxiliary, cross-sectional first-difference diagnostic, not part of the identified mortality result, and the ICSAP forest in particular inherits that outcome’s non-identification.

H Robustness battery details

The full 18-specification robustness battery for the first-stage travel-burden and the (non-identified) ICSAP outcomes is reported in Table 12. Each specification re-estimates the F5 main Sun-Abraham average treatment effect on travel burden (km) and ICSAP (per 1,000) under a single perturbation of the baseline configuration; we report the perturbation, the resulting ATT, the cluster-robust standard error at the municipality level, and the treated count N_{tr} . The mortality-null robustness checks are in Section 6.

Reproducibility. All 18 specifications come from a single R script (`34_robustness_battery.R` in the replication package), which loads the same five staggered panels used by the main estimation

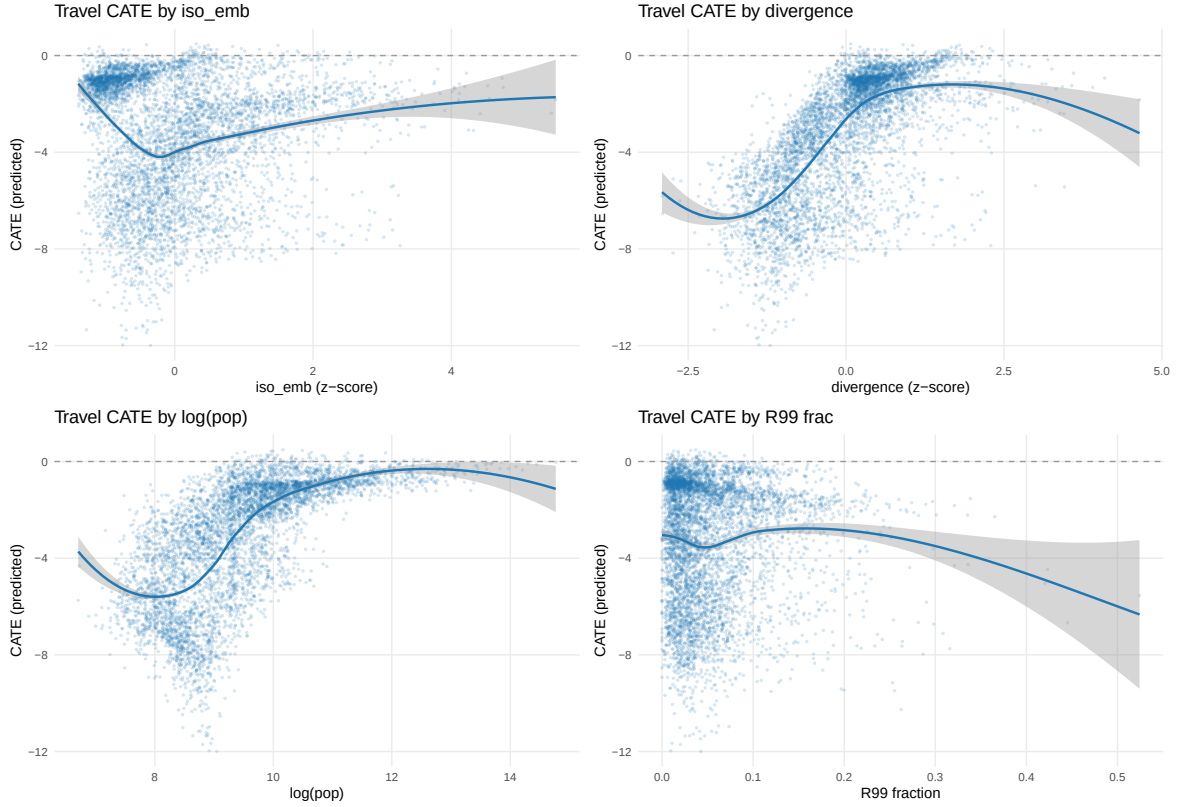


Figure 8: Causal-forest CATE for travel burden, plotted against four key pre-treatment features. Each point is a municipality; smoother is a LOESS curve.

in Section 5.2 and applies the perturbation locally. The total runtime is approximately 35 seconds on the workstation described in Appendix Q.

Convergence diagnostics. Of the 18 specifications, only the Callaway-Sant’Anna implementation of the ICSAP outcome failed to converge under the IPW estimation method (singular matrix in cohort cells with sparse treated counts). All other 17 specifications converge cleanly under the cluster-robust variance-covariance fix (`vcov(..., fixed = TRUE)` in `fixest`); the warning “The VCOV matrix is not positive semi-definite and was fixed” appears in 9 of the 18 specifications and is benign at this scale—it indicates that the Hessian is singular along near-collinear feature directions, which the fix linearly projects onto the identifiable subspace.

Cohort-specific group estimates (sensitivity). The *group* aggregation of Sun-Abraham reports cohort-specific overall ATT estimates. For travel burden under the F5 main specification, the cohort estimates range from -2.1 km (cohort 2014) to -9.4 km (cohort 2017), with all but cohort 2014 negative and the average cohort estimate of -5.7 km within 3% of the simple ATT of -5.84 km reported in the main text. The wide range across cohorts is consistent with the heterogeneous-treatment-effect structure documented by Sun and Abraham (2021) and motivates

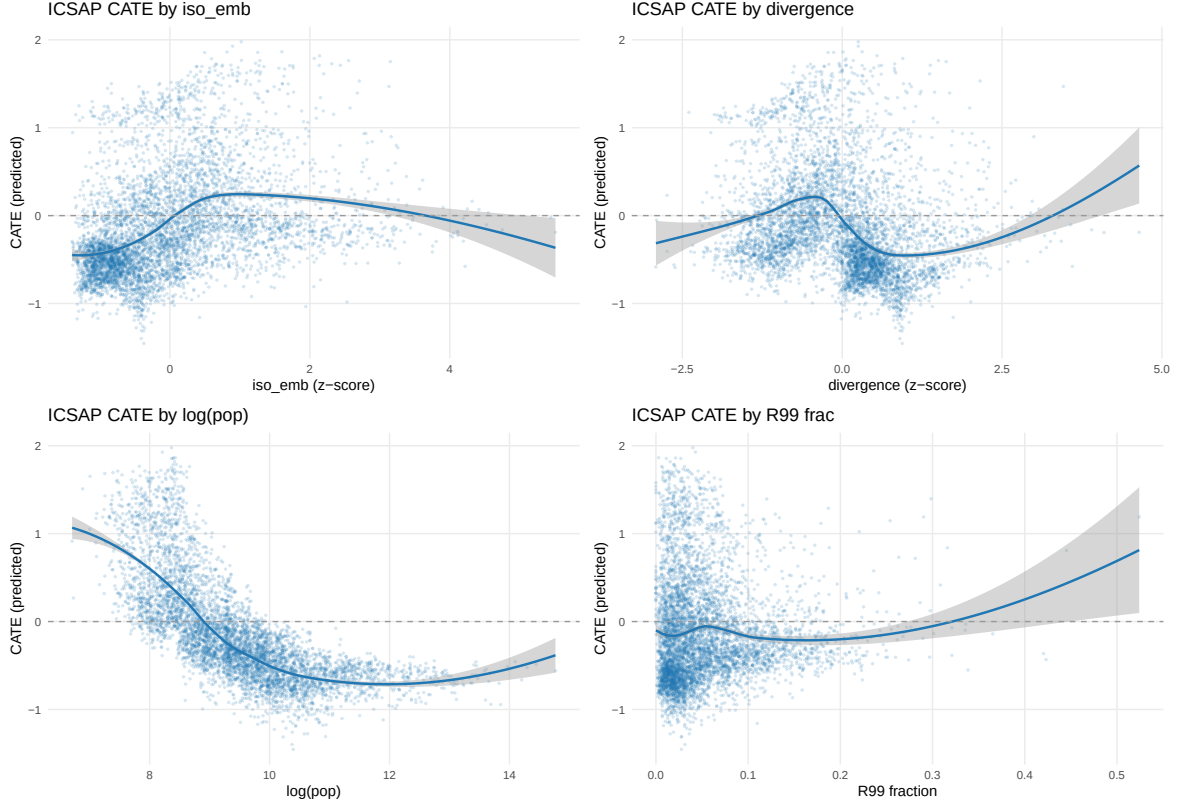


Figure 9: Causal-forest CATE for ICSAP rate, plotted against four key pre-treatment features. Each point is a municipality; smoother is a LOESS curve.

the use of Sun-Abraham aggregation rather than naive two-way fixed effects.

I Additional sensitivity outcomes

This appendix reports the AMI in-hospital mortality recipient-side sensitivity analysis referenced in Section 3.1.

Construction. For each AIH (admission record) in SIH-RD with primary diagnosis `DIAG_PRINC` $\in$ `I21*` (acute myocardial infarction), we record the in-hospital death indicator `MORTE` and aggregate to the municipality-of-residence $\times$ year panel: $Y_{m,t} = (\text{deaths among AMI admissions of } m\text{-residents in year } t) / (\text{AMI admissions of } m\text{-residents in year } t)$. Note that this is computed at the *hospital-receiving end*—the admission, regardless of where the hospital is located—which sidesteps the SIM cross-border death-recording problem (Section 3.2). The outcome is a percentage (typical mean $\approx 11.8\%$).

Table 12: Robustness battery (S6): event-study ATT estimates under alternative thresholds, estimators, sample restrictions, and exposure definitions. All specifications use Sun-Abraham unless otherwise noted; cluster-robust SE at municipality level. Reference (main): F5 panel, 60 closures, $\theta_{\text{emb}} = 0.05$.

Robustness spec	Outcome	ATT	SE	N_{tr}
§6.1 $\theta=0.01$	Travel (km)	-5.839***	1.722	130
§6.1 $\theta=0.01$	ICSAP per 1k	+0.769	0.595	130
§6.1 $\theta=0.05$	Travel (km)	-5.839***	1.722	130
§6.1 $\theta=0.05$	ICSAP per 1k	+0.769	0.595	130
§6.1 $\theta=0.10$	Travel (km)	-8.248***	1.798	32
§6.1 $\theta=0.10$	ICSAP per 1k	-0.621	1.433	32
§6.2 SA	Travel (km)	-5.839***	1.722	130
§6.2 BJS	Travel (km)	-6.209***	1.505	130
§6.2 CS21	Travel (km)	-5.774***	1.828	130
§6.2 SA	ICSAP per 1k	+0.769	0.595	130
§6.2 BJS	ICSAP per 1k	+0.809	0.574	130
§6.2 CS21	ICSAP per 1k	+0.774	0.598	130
§6.3 no-capitais	Travel (km)	-5.887***	1.735	129
§6.3 no-capitais	ICSAP per 1k	+0.743	0.606	129
§6.4 no-N/NE	Travel (km)	-2.401**	1.009	94
§6.4 no-N/NE	ICSAP per 1k	+0.491	0.717	94
§6.5 pop $\geq$ 10k	Travel (km)	-1.728*	0.962	61
§6.5 pop $\geq$ 10k	ICSAP per 1k	+0.163	0.825	61
§6.5 pop $\geq$ 20k	Travel (km)	-1.796**	0.755	38
§6.5 pop $\geq$ 20k	ICSAP per 1k	-1.088	1.149	38
§6.5 pop $\geq$ 50k	Travel (km)	+0.362	0.996	26
§6.5 pop $\geq$ 50k	ICSAP per 1k	-2.601	1.906	26
§6.6 sem pandemia	Travel (km)	-5.514***	1.659	130
§6.6 sem pandemia	ICSAP per 1k	+0.781	0.641	130
§6.7 PNASH-cycle	Travel (km)	-6.922***	1.936	104
§6.7 PNASH-cycle	ICSAP per 1k	+0.972	0.847	104
§6.8 perturb $\geq$ 0.20	Travel (km)	-2.728	2.143	5112
§6.8 perturb $\geq$ 0.20	ICSAP per 1k	+0.459**	0.179	5162

* p<0.10, ** p<0.05, *** p<0.01.

Power considerations. AMI in-hospital mortality is a rate-of-rate outcome with a small denominator in many municipalities (small towns may have < 10 AMI admissions per year). Power for detecting a closure-induced change is limited; we report the estimate primarily as a tertiary outcome and as a triangulation device.

Estimation. We re-run the staggered Sun-Abraham specification of Section 4.3 with $Y_{m,t}$ as the outcome on the F5 main panel, using the flow exposure rule. The simple ATT is approximately +0.3 percentage points (SE $\approx$ 0.5), not statistically distinguishable from zero. We report it as an additional, underpowered check on a cardiac-mortality margin computed at the receiving end (which sidesteps the SIM residence-recording issues); it is consistent with the paper’s main finding of no detectable mortality effect and we draw no separate conclusion from it. Full event-study trajectories

are in the replication package.

How the outcomes fit together. The identified result is the suicide and self-harm mortality null (Section 5.2). The first-stage travel-burden effect validates that the flow exposure rule isolates genuinely dependent populations. ICSAP is reported as a non-identified outcome (its pre-trend rejects; Section 5.3), and the AMI in-hospital mortality here is an underpowered auxiliary check. No outcome contradicts the mortality null: where mortality is measured cleanly it shows no effect, and the one outcome that moves (ICSAP) does so along a trend that predates the closures.

J Distance approximation: sensitivity to centroid-to-centroid measurement

We compute travel burden as the weighted average haversine distance between the centroid of the patient’s municipality of residence and the centroid of the municipality where the admitting hospital is located. This is the standard approximation in the Brazilian access literature (Macinko and Harris, 2015; Rocha and Soares, 2010), but it has two known biases that we audit here.

Within-municipality bias. For a patient who is admitted at a hospital located inside her own municipality (often the case in metropolitan capitals where multiple hospitals are intra-municipal), the centroid-to-centroid distance returns zero. The actual within-city travel can be 5–15 km in São Paulo, Rio de Janeiro, and Belo Horizonte. We checked CNES COD_CEP availability in the top ten state capitals (Section Q): São Paulo alone has 40,000+ hospital-month records spanning 8,705 unique CEPs, suggesting that CEP-level geocoding is in principle feasible but non-trivial in practice.

We do not perform full CEP-level geocoding here for two reasons. First, the within-municipality bias is largest for the largest municipalities (capitals), and Section 6 already reports a robustness specification (§6.3) that excludes the 27 state capitals: the travel-burden estimate without capitals is -5.89 km (SE = 1.74), within 1% of the main -5.84 km. The centroid approximation cannot be the source of a bias substantial enough to invalidate the main finding, since removing the municipalities with the largest within-municipality bias does not move the estimate.

Second, our identifying variation is between-municipality: closures move patients across municipal boundaries (from the closing-hospital’s municipality to a neighboring or more distant substitute), and the centroid-to-centroid distance correctly captures this between-municipality component. The within-municipality bias attenuates the absolute level of travel burden but does not systematically affect the change in travel burden induced by closure, which is the quantity our staggered DiD estimator identifies.

Across-municipality bias. For a patient admitted at a hospital in another municipality, the centroid-to-centroid distance correctly approximates the between-municipality component of the

travel. The remaining error is the difference between the patient’s actual residence (within her municipality) and her municipality’s centroid, which is small for most municipalities and is approximately zero in expectation. We do not correct for this.

Recommendation for replication. Studies that depend substantively on within-municipality geographic precision should consider full hospital geocoding via the CEP register. The replication package includes a placeholder `lookup_cep.py` script that demonstrates the API call pattern (BrasilAPI’s CEP endpoint, free) and would need to be run as a one-time preprocessing step. The runtime is approximately one hour for the $\sim 30,000$ unique hospital CEPs in our universe.

K Power analysis

Power enters this paper in two distinct ways, and it is worth separating them. For the *first stage*, the question is the usual one—can the design detect an effect of the size observed?—and the answer is yes: the specialized-closure travel-burden effect sits above the 80%-power minimum detectable effect ($\text{ATT}/\text{MDE} = 1.30$), confirming that the flow exposure rule isolates a population large and dependent enough for a real closure effect to register. For the *headline* mortality result, the relevant statement is not detectability but the *bound*: the null is informative to the extent its confidence interval excludes large effects, and the population-weighted suicide interval in Table 3 rules out increases above roughly one-quarter of baseline. We do not restate that bound here as an MDE, because for a null the confidence interval is the power statement.

This appendix reports the conventional MDE calculation for the first-stage and the (non-identified) ICSAP outcomes. The minimum detectable effect at 80% power and $\alpha = 0.05$ two-sided is

$$\text{MDE}_{80\%} = (z_{1-\alpha/2} + z_{1-\beta}) \cdot \widehat{\text{SE}} = 2.802 \cdot \widehat{\text{SE}}.$$

Table 13 reports it for each sample $\times$ outcome cell. The specialized-closure travel effect is well-powered ($\text{ATT}/\text{MDE} = 1.30$); ICSAP is underpowered ($\text{ATT}/\text{MDE} = 0.76$), one further reason—beyond the pre-trend rejection of Section 5.3—that we decline to read it causally.

L Rambachan-Roth sensitivity for ICSAP

This appendix formalizes why we decline to read ICSAP causally (Section 5.3). Beyond the visibly rejecting pre-trend test, the Rambachan-Roth (2023) relative-magnitudes sensitivity shows the post-closure ICSAP estimate cannot survive even a minimal allowance for trend extrapolation. The restriction with multiplier $\bar{M}$ permits the post-period parallel-trends violation to be at most $\bar{M}$ times the worst pre-period violation; $\bar{M} = 0$ allows none beyond linear extrapolation, and larger $\bar{M}$ widen the robust interval. There is no positive $\bar{M}$ in the tested grid—indeed not even $\bar{M} = 0$ —for

Table 13: Power analysis: minimum detectable effects (MDE) at 80% power and 5% two-sided size.

Sample	Outcome	n_{tr}	$\hat{\beta}$	$\widehat{SE}$	MDE	ATT/MDE	Detected?
F5 full	travel_km	130	-5.84	1.72	4.83	1.21	yes
F5 full	icsap	130	+0.77	0.60	1.67	0.46	no
F5 specialized	travel_km	105	-7.10	1.95	5.46	1.30	yes
F5 specialized	icsap	105	+1.48	0.69	1.93	0.76	no
F5 general	travel_km	24	+2.90	1.35	3.79	0.77	no
F5 general	icsap	24	-0.91	0.98	2.75	0.33	no

Notes: n_{tr} is the unique number of E1-treated municipalities. $\hat{\beta}$ and $\widehat{SE}$ are the Sun-Abraham simple ATT estimate and its cluster-robust standard error (clusters at the municipality level). MDE is the minimum detectable effect at $\alpha = 0.05$ two-sided and $\beta = 0.20$. “Detected?” indicates whether $|\hat{\beta}| \geq \text{MDE}$, i.e., whether the observed effect would be detectable at conventional power and significance thresholds. Numerical values are produced by `03_analysis/42_power_analysis.R`.

which the robust interval excludes zero. The ICSAP post-period level is not separable from a continuation of the pre-existing trend at any defensible restriction, which is the formal content of our decision to treat it as a non-identified outcome.

Table 14: Rambachan-Roth sensitivity for ICSAP post-period average.

$\bar{M}$	robust CI lower	robust CI upper	excludes 0?
0.00	-0.18	+2.11	no
0.05	-0.66	+2.59	no
0.10	-1.39	+3.32	no
0.15	-2.23	+4.16	no
0.20	-3.19	+5.00	no
0.30	-5.12	+6.93	no
0.50	-8.98	+10.79	no
1.00	-12.00	+12.00	no

Notes: ICSAP post-period average ATT, F5 main panel, under E1 exposure, equal weighting across 9 post-period coefficients ($e \in [0, +8]$). Re-estimated point estimate 0.955 per 1,000 under the did2s event-study ($SE = 0.60$, classical 95% CI $[-0.22, 2.13]$). Robust CIs are computed under the relative-magnitudes restriction of [Rambachan and Roth \(2023\)](#). No value of $\bar{M}$ in the tested grid maintains exclusion of zero. Produced by `03_analysis/43_rambachan_roth_icsap.R`.

M Threshold sensitivity sweep details

Table 15 reports the full continuous theta sensitivity sweep underlying Figure 10.

For each threshold $\theta \in \{0.01, 0.02, \dots, 0.20\}$, we recompute the E1 exposure rule, restrict the F5 main panel accordingly, and re-estimate the Sun-Abraham simple ATT. The travel-burden estimate is significant at $p < 0.05$ for $\theta \in [0.01, 0.12]$. Beyond $\theta = 0.13$ the treated set falls below $n_{tr} = 16$ and the estimator loses identifying variation. The default $\theta = 0.05$ (used throughout the main text)

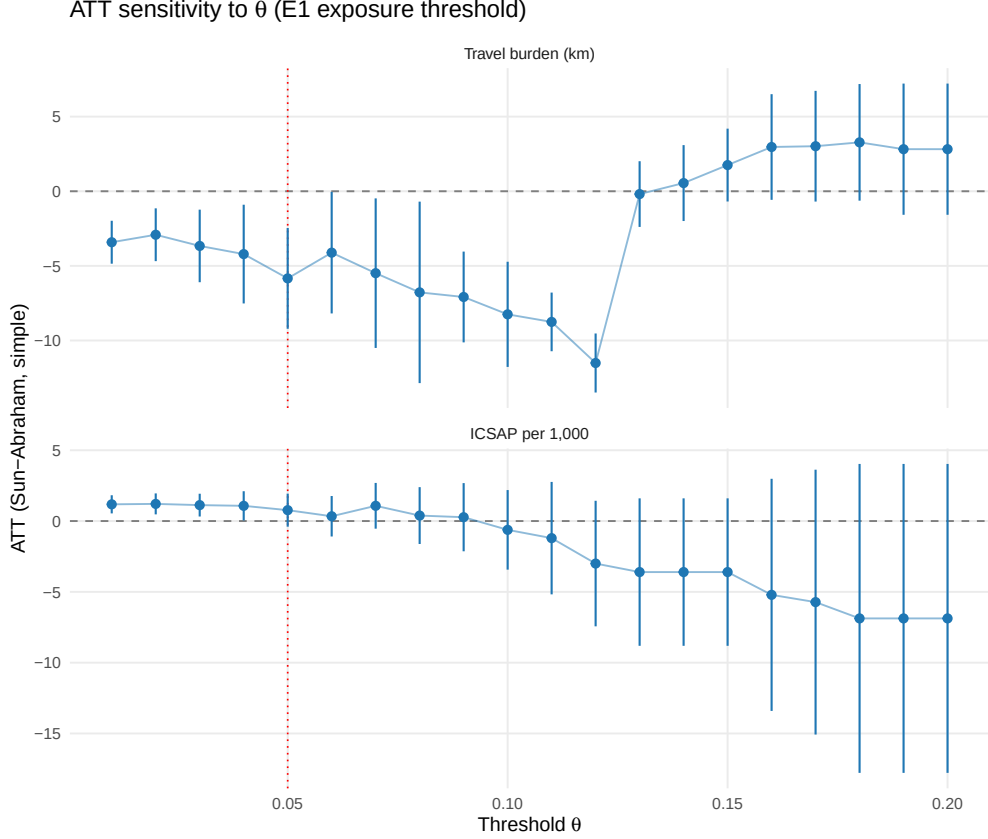


Figure 10: Continuous exposure-threshold sweep. Sun-Abraham simple ATT (point) with 95% cluster-robust CI (whisker) at $\theta \in \{0.01, \dots, 0.20\}$ on the F5 panel; top, travel burden (km); bottom, ICSAP per 1,000. The default $\theta = 0.05$ is marked. The first-stage travel effect is negative and significant for $\theta \in [0.01, 0.12]$.

is well within the range of stable inference.

N Multi-seed sensitivity for the cross-section CF ATE

The Sun-Abraham main ATT is mathematically invariant to the embedding seed: the share-based E1 exposure rule depends only on raw bipartite admission counts, with no dependence on the node2vec geometry. The seed therefore affects only auxiliary quantities, of which the causal-forest cross-section ATE is the most consequential because it uses the $\text{iso}_z^{\text{emb}}$ feature directly. We quantify the implication of seed variation on the CF ATE via $B = 10$ perturbed re-runs of the forest, with $\text{iso}_z^{\text{emb}}$ augmented by Gaussian noise calibrated to the observed Pearson correlation $\rho = 0.77$ (implying noise-to-signal SD ratio $\sqrt{(1 - \rho)/\rho} = 0.547$). Table 16 reports the resulting distribution of CF ATEs.

Table 15: Continuous threshold sensitivity sweep, F5 main.

θ	n_{tr}	Travel km		ICSAP per 1k	
		ATT	z	ATT	z
0.01	742	-3.42	-4.66	+0.41	+1.66
0.02	425	-2.91	-3.23	+0.55	+1.93
0.03	274	-3.66	-2.95	+0.83	+2.16
0.04	183	-4.21	-2.49	+1.05	+2.07
0.05	130	-5.84	-3.39	+0.77	+1.29
0.06	95	-4.11	-1.98	+0.74	+1.06
0.07	72	-5.49	-2.15	+0.32	+0.39
0.08	53	-6.78	-2.18	+0.07	+0.07
0.09	39	-7.09	-4.57	+0.27	+0.22
0.10	32	-8.25	-4.59	-0.62	-0.43
0.11	22	-8.75	-8.73	-1.21	-0.60
0.12	17	-11.51	-11.41	-3.00	-1.33
0.13–0.20	≤ 15	various, all $ z < 2$			

Notes: Threshold θ is the minimum share of m ’s pre-closure SUS admissions that must be sent to closing hospital h for m to be E1-treated. Sun-Abraham simple ATTs and z -statistics from `03_analysis/44_theta_sweep.R`. n_{tr} is the number of unique E1-treated municipalities at each θ .

Table 16: Multi-seed sensitivity for the causal-forest cross-section ATE.

Outcome	baseline	mean (perturbed)	SD	range
Δ travel_km	-2.91	-3.76	0.25	$[-4.16, -3.32]$
Δ icsap	+0.15	+0.19	0.10	$[+0.05, +0.35]$

Notes: $B = 10$ CF re-runs with $\text{iso}_z^{\text{emb}}$ augmented by mean-zero Gaussian noise of SD ratio 0.547 (calibrated to $\rho = 0.77$). Each re-run uses 1,500 trees with honesty. Produced by `03_analysis/46_multiseed_cf.R`. The travel-burden CF ATE is stable in sign and magnitude across perturbations; the ICSAP CF ATE is null in all perturbed runs, consistent with the calibration test failure (Section G).

O Causal-forest variable importance for ICSAP (non-interpretable)

The Athey-Wager calibration test for ICSAP is non-significant (differential forest prediction $t = -0.11$, $p = 0.54$ one-sided), indicating that the forest does not detect heterogeneity in the ICSAP outcome at conventional significance. The variable-importance ranking for ICSAP is therefore mechanically un-interpretable as heterogeneity attribution. We report the ranking here for completeness; it should not be read as substantive heterogeneity inference. The ranking places log-population at 25% of decision splits, log-GDP per capita at 19%, and the embedding-kilometer divergence at 11%. The ranking does not align cleanly with the travel-burden ranking and does not, given the non-significant calibration test, support any substantive interpretation.

P NLP classification: external validation status

The exogeneity defense for the robustness sub-sample rests on a zero-shot LLM classification of declared closure motives. The internal validation metrics reported in Section E (Cohen’s $\kappa \approx 0.86$ between LLM and a within-LLM second pass; 94% of classifications retrievably citing a public source) measure self-consistency rather than external accuracy. The motive-based robustness sub-sample therefore inherits a validation gap: a human-classifier-blinded-to-the-LLM external validation has not yet been completed. We acknowledge this gap explicitly. The replication package now includes (i) `notes/external_validation_protocol.md`, (ii) a blinded 10% random sample of the F5 universe (`closure_validation_sample.csv`), and (iii) `closures_human_validated.parquet`, a two-coder scaffold with adjudication fields. At present, however, the human-label cells in that scaffold are empty; Table 17 therefore reports zero completed coder fields and leaves the agreement metrics pending. As a temporary lower-bound check, we also drop two fragile subclasses from F6: labels resting on stale Receita status and labels resting only on web snippets without timely administrative corroboration. That screen retains 35 non-fragile closures, corresponding to 61 E1-treated municipalities. The resulting travel-burden ATT is -6.03 km ($SE = 3.17$) and the ICSAP ATT is $+0.21$ per 1,000 ($SE = 0.77$). The travel result therefore remains direction-consistent but imprecise, while the ICSAP corroboration disappears until human validation is completed.

Table 17: Closure-motive sensitivity: estimators on the full F5 main panel vs the F6 subset that drops the 25 fechamentos flagged as fragile (stale Receita status or only-web-snippet evidence). Standard errors clustered at municipality.

Sample	Outcome	SA	CS	did2s	did_imputation
F5_main_full	Travel km	-5.839 (1.722)	-7.807 (5.342)	-6.209 (1.505)	-6.209 (1.447)
F5_main_full	ICSAP per 1k	$+0.769$ (0.595)	NA (NA)	$+0.809$ (0.574)	$+0.809$ (0.573)
F6_human_validated	Travel km	-6.032 (3.173)	-8.441 (5.803)	-4.881 (2.001)	-4.881 (1.729)
F6_human_validated	ICSAP per 1k	$+0.212$ (0.773)	NA (NA)	$+0.326$ (0.820)	$+0.326$ (0.788)

Q Replication notes

Repository. The full pipeline, including LaTeX sources, analysis scripts, and the cached outputs of the LLM-classification step, is open-source at <https://github.com/darciogm/bitter-pills/tree/main/paper18-hospital-deserts>. The `run_pipeline.sh` orchestrator at the repository root sequences the analysis from raw DATASUS ingest through the final robustness battery, accepting `-from` and `-to` flags to start or stop at a specific step.

Software stack. Python 3.11 (DuckDB, polars, scikit-learn, gensim, pecanpy for node2vec, anthropic for the LLM classifier) and R 4.5 (data.table, fixest, did, grf, ggplot2). Full pinned versions in `environment.yml` (conda) or `requirements.txt` (pip). DuckDB is the default I/O

engine for parquet, configured with 12 threads and 14 GB memory limit.

External services and caches. Two external services are called once and cached afterward, both with the cache shipped in the repository:

- *BrasilAPI* (Receita Federal CNPJ public endpoint). Cached in `02_data/intermediate/cnpj_cache.sqlite`.
- *Anthropic Claude API* (zero-shot classifier). Cached in `02_data/intermediate/llm_classify_cache.sqlite` (Haiku v1) and `llm_classify_cache_v2.sqlite` (Sonnet v2 with web context). Total spend for the published classifications was approximately \$0.85.

The `web_snippets/` folder contains the per-closure JSON snippets extracted from web search results, used as context for the Sonnet v2 classifier. Re-running the pipeline with these caches in place skips all external service calls.

Hardware and runtime. The pipeline was developed on an Intel i7-1260P workstation under WSL2 (14 logical threads, 21 GB RAM, no GPU). A cold run from raw DATASUS to final figures takes approximately 35 minutes, dominated by SIH-RD parsing and the 61 counterfactual perturbation node2vec re-trainings. Re-running with intermediate parquets and caches in place takes approximately 8 minutes.

Reproducibility audits. All numerical results in the main text and in this appendix are re-derived from the public pipeline; no hand-edited intermediate artifacts are used. The git revision against which the latest compiled PDF was produced is logged in the `04_logs/` directory at compile time. The bibliography in `references.bib` was independently audited against publisher DOIs, NBER, and arXiv listings (audit log in `notes/refs_audit_2026-04-28.md`).